

Asset Pricing under Knightian Uncertainty: A Case for Optimism*

Soroush Ghazi

Mark Schneider

July 1, 2022

Abstract

We apply a representative agent model separating beliefs, Knightian uncertainty, and uncertainty attitude, to empirical puzzles for the aggregate stock market. Allowing for optimistic uncertainty attitudes helps match the risk-free rate, equity premium, variance risk premium, and risk-neutral entropy, and produces the observed pricing kernel U-shape. The low (high) persistence of market uncertainty (market optimism) generates return predictability of the price-dividend ratio, variance risk premium, market crash probability, and market correlation and helps explain short-run (long-run) dynamics of the pricing kernel. Accounting for the interaction of market volatility, uncertainty, and optimism helps identify the risk-return tradeoff for the aggregate market.

JEL Classification: D8, G40, G41

Keywords: Optimism, Uncertainty, Skewness, NEO-EU CAPM, Stock Market Anomalies

*We thank Aurelien Baillon, Turan Bali, Geert Bekaert, Sylvain Chassang, Lauren Cohen, Bob Day, Stephen Dimmock, Sebastian Ebert, Simon Grant, Yehuda Izhakian, Tigran Melkonyan, Scott Murray, Sugata Ray, Mike Shor, Tobias Sichert, David Schreindorfer, Robert Shiller, Julian Thimme, Laura Veldkamp, Jeffrey Wurgler, Alexander Zimmer, and seminar participants at the University of Connecticut, the 2020 SEA Conference, the 2021 North American Summer Meeting of the Econometric Society, and the 2021 China Meeting of the Econometric Society for helpful feedback regarding this research. Affiliation of authors: Culverhouse College of Business, University of Alabama, 361 Stadium Drive, Tuscaloosa, AL 35487. Emails: Ghazi: sghazi@cba.ua.edu; Schneider: maschneider4@cba.ua.edu.

1. Introduction

Motivated by empirical challenges for the consumption capital asset pricing model (CCAPM) based on an expected utility (EU) representative agent (*e.g.*, Mehra and Prescott, 1985), models of uncertainty aversion have been applied to financial markets with promising results, matching basic stock market moments such as the large average return of stocks over bonds while maintaining a low risk-free rate.¹ Models of uncertainty aversion such as the maxmin multiple priors framework of Gilboa and Schmeidler (1989), the robust control approach of Hansen and Sargent (2001), and the smooth model of Klibanoff et al. (2005) are inspired by the distinction between choices involving risk (in which probabilities of events are known) and Knightian uncertainty (Knight, 1921) in which agents lack the information or confidence to assign unique probabilities to events. These major frameworks applied to asset pricing assume that agents are uniformly averse to Knightian uncertainty and overweight the possible negative outcomes relative to an EU agent.²

As demonstrated by experiments, however, (*e.g.*, Baillon and Bleichrodt, 2015; Baillon and Emirmahmutoglu, 2018; Dimmock et al., 2015; Kocher et al., 2018; Trautmann and Van De Kuilen, 2015), uncertainty aversion is not universal. Many individuals exhibit both an aversion to uncertainty over left-tail outcomes and a positive disposition toward uncertainty for right-tail outcomes.³ As the major frameworks for choice under Knightian uncertainty applied to asset pricing emphasize only uncertainty aversion, a natural question is whether the attraction toward uncertainty documented for low-likelihood lottery-like payoffs in experiments manifests itself in the aggregate market. If so, models which allow for optimism toward uncertainty may capture phenomena that have not been demonstrated under uniform uncertainty aversion.

Today, the α -maxmin and Choquet expected utility models (Ghirardato et al., 2004; Schmeidler, 1989) are two other main paradigms for modeling uncertainty preferences. Non-extreme outcome expected utility (NEO-EU) preferences (Chateauneuf et al., 2007) satisfy the axioms of both models and achieve a three-way separation between an agent's beliefs (represented by a unique additive probability measure), the agent's perceived uncertainty (ranging from objectively known probabilities to complete uncertainty), and the agent's uncertainty attitude (ranging from extreme pessimism to extreme optimism).⁴ The NEO-EU model captures a preference for robustness to uncertainty (by overweighting the agent's perceived worst-case scenario) and a preference for positive skewness (by overweighting the perceived best-case scenario), thereby accounting for two of the main empirical limitations of EU (*e.g.*, Dimmock et al., 2015; Ebert and Karehnke, 2021).⁵

¹Guidolin and Rinaldi (2013) reviews the early literature on applying Knightian uncertainty to asset pricing models.

²Since Ellsberg (1961), such 'uncertainty aversion' is also referred to as ambiguity aversion. Following Schmeidler (1989), we use the term uncertainty aversion. We also refer more generally to uncertainty attitudes which may reflect degrees of aversion or attraction to Knightian uncertainty.

³In financial markets, this behavior can be viewed as a preference for hedging uncertainty over left-tail outcomes (which might reduce anxiety over how much can be lost), and a preference for speculating on uncertainty over right-tail outcomes (which might increase excitement over how much can be gained).

⁴Chateauneuf et al. (2007) refer to this model as neo-additive expected utility since it is additive in non-extreme outcomes.

⁵In a large representative sample, Dimmock et al. (2015) and Dimmock et al. (2016) find that many people

While the α -maxmin model has also been studied in market experiments (Bossaerts et al., 2010), its predictions regarding aggregate stock market prices remain largely unexplored.⁶

The present paper investigates the effects of Knightian uncertainty and uncertainty attitude on the aggregate stock market and makes three main contributions: First, we calibrate a generalization of the CCAPM with a type of α -maxmin representative agent. The calibration shows that the equity premium puzzle (Mehra and Prescott, 1985), the variance premium puzzle (Zhou, 2018), and the VIX-SVIX puzzle (Martin, 2017) can be explained simultaneously with a small degree of optimism toward uncertainty and a small deviation from EU. Second, we apply the model to obtain a unified explanation for the ability of the variance risk premium (VRP), the market crash probability (CP), the market implied correlation (IC), the price-dividend ratio (PD), and the shape of the stochastic discount factor, to predict future market returns. This explanation makes the new empirical prediction that return predictability of the VRP, market crash risk, market correlation, and the right tail of the stochastic discount factor is concentrated in periods in which PD is low. We find strong empirical support for this prediction and show that it is distinct from return predictability originating from recessions. We further show that the same property of uncertainty and optimism that drives return predictability also predicts systematic time variation in the shape of the stochastic discount factor or ‘pricing kernel’, thereby providing a more complete explanation for the pricing kernel puzzle (Cuesdeanu and Jackwerth, 2018). Third, we derive an approximation for the market equity premium which directly relates the equity premium to market volatility, market uncertainty, and market optimism. The formula consists of a risk-based component (the product of market volatility and market uncertainty) and a speculation component (the product of market volatility, market uncertainty, and market optimism). We find that including lagged versions of both components in the return predictability regression is needed to identify the risk-return tradeoff for the aggregate market and predict the equity premium.

Our paper contributes to four basic types of empirical puzzles that challenge the CCAPM as well as more recent asset pricing models. The first type pertains to matching basic stock market moments. The equity premium puzzle (Mehra and Prescott, 1985) highlights the difficulty of matching the large historical excess returns of stocks over bonds while generating a low risk-free rate under the standard CCAPM based on an expected utility-maximizing representative agent. Mehra and Prescott (1985) show that even with a risk aversion coefficient of 10, the CCAPM generates an equity premium of 0.35%, far below their estimate of the historical equity premium above 6%. The variance premium puzzle (Zhou, 2018) highlights the difficulty of the CCAPM and subsequent representative agent asset pricing models of matching the historical difference between the risk-neutral and the physical market variance. The VIX-SVIX puzzle is the finding that major

have non-neutral attitudes toward uncertainty which are inconsistent with EU. Ebert and Karehnke (2021) show analytically that the preferences toward skewness observed in laboratory experiments cannot be explained by EU, but are well-captured by models of probability weighting. Holzmeister et al. (2020) find that for both finance professionals and the general population, risk perception is determined by the skewness of payoffs rather than by volatility.

⁶The exception is the special case with no optimism which corresponds to the maxmin multiple priors model (Gilboa and Schmeidler, 1989).

macro-finance models (*e.g.*, Campbell and Cochrane, 1999; Bansal and Yaron, 2004; Wachter, 2013) which overweight only the left tail of the risk-neutral distribution have difficulty matching the unconditional difference between the risk-neutral entropy (VIX) and the risk-neutral variance of the excess market return (SVIX), being off by an order of magnitude (Martin, 2017; Bekaert et al., 2020). Overweighting only the left tail of the risk-neutral distribution is also characteristic of the major uncertainty preference frameworks applied to asset pricing (Gilboa and Schmeidler, 1989; Klibanoff et al., 2005; Hansen and Sargent, 2001). However, as entropy is more sensitive to the left tail of the distribution in comparison to the variance, the difference between VIX and SVIX reveals that the left tail of the risk-neutral distribution is only moderately heavier than the right tail.

The second type of puzzle pertains to the ability of certain market indicators to predict future stock market returns. In particular, PD has been shown to negatively predict future aggregate stock market returns (Campbell and Shiller, 1988), while the VRP (Bollerslev et al., 2009), market crash probability (Martin, 2017), and market-implied correlation between stocks (Buss et al., 2019) each positively predict market returns. The third empirical challenge to the CCAPM, the pricing kernel puzzle, refers to the now standard empirical finding that the stochastic discount factor is U-shaped when projected onto returns (Bakshi et al., 2010; Sichert, 2020), while traditional asset pricing models predict the stochastic discount factor to be monotonically decreasing, consistent with a purely risk-averse representative agent. The fourth type of basic puzzle (Yu and Yuan, 2011) is the empirical difficulty in identifying a positive risk-return tradeoff for the aggregate market (higher expected returns in times of higher stock market volatility).⁷

To elaborate on our second contribution, we impose only one general assumption on the dynamics of market uncertainty and market optimism: Market uncertainty follows a transitory stochastic process whereas market optimism follows a persistent process. Empirically, our measures of market uncertainty (such as the VRP) have low persistence, while our measures of market optimism (such as the Baker and Wurgler (2006) market sentiment (BW) index) have high persistence.⁸ Theoretically, the low persistence of market uncertainty arises naturally if uncertainty changes (beliefs are updated and the uncertainty is resolved) in response to streams of new information each period. The high persistence of market optimism (reflecting the market's uncertainty attitude) depends on more stable traits like investors' education, age, or other demographic factors.⁹ These traits are not updated as frequently in individuals. However, in the aggregate (for a representative agent), they do respond to slow-moving environment variables.¹⁰ The high persistence of market sentiment is

⁷The literature on the risk-return tradeoff is synthesized by Yu and Yuan (2011) and Brenner and Izhakian (2018).

⁸For example, the VRP constructed based on Martin (2017) has a lag 1 auto-correlation of 0.55, and the VRP constructed from the VIX index has a lag 1 auto-correlation of 0.62. In contrast, for our sample period the Baker and Wurgler (2006) index has a lag 1 auto-correlation of 0.97, and the Huang et al. (2015) sentiment index constructed from the same sentiment proxies used in Baker and Wurgler (2006) has a lag 1 auto-correlation of 0.99.

⁹For instance, Grevenbrock et al. (2020) find that a person's degree of optimism or pessimism is related to age and that it changes systematically over the life-cycle.

¹⁰Even periods of extreme optimism, such as the dot-com bubble may be better explained by changes in the demographic composition of investors than by large changes in optimism within investors. For instance, Greenwood and Nagel (2009) find that "around the peak of the technology bubble, mutual funds run by younger managers are more heavily invested in technology stocks, relative to their style benchmarks, than their older colleagues. Furthermore, young managers, but not old managers, exhibit trendchasing behavior in their technology stock investments."

also noted in prior work. For instance, Schmeling (2007) comments, “the building up of excessive optimism or pessimism, *i.e.*, sentiment, is likely to be a persistent process whose effects on stock prices would be hard to detect over short horizons of one or two months” (p. 128). Schmeling then finds empirically, “all sentiment series are highly persistent” (p.131).

The high (low) persistence of market optimism (uncertainty) leads to a new mechanism for stock return predictability. To provide intuition for this mechanism, we start with the model implication that a low PD and high VRP, CP, or IC, on average correspond to periods of low optimism and high uncertainty (we demonstrate these implications in Section 4). Suppose that the representative agent’s uncertainty attitude is persistent, but uncertainty is transitory. Consider a period in which PD is low, and hence (on average) optimism is low and uncertainty is high. Then, in the near future, optimism is still low (due to its high persistence) but the high uncertainty is mostly resolved (due to its transitory nature). As uncertainty resolves in the near future (market uncertainty decreases), the future PD and hence, the future return increases, generating positive return predictability for VRP, CP, and IC in periods when PD is low. In the model, this predictability comes primarily from low PD periods since when market optimism, α , is high (and so, on average, PD is high), the effect of uncertainty is mitigated.

The low persistence of market uncertainty along with the high persistence of market optimism also helps to explain the dynamics of the pricing kernel. The pricing kernel puzzle is the finding that the stochastic discount factor is not monotonically decreasing when projected onto returns but instead increases at the right tail of the return distribution, generating a U-shape (Bakshi et al., 2010; Sichert, 2020). The U-shape is also a feature of the stochastic discount factor in our setting, with the increasing part of the U-shape arising from the positive skewness preference of the representative agent, and it is increasing in both market uncertainty and market optimism. Our approach goes beyond explaining the shape of the pricing kernel by also making and testing a new prediction regarding its time variation. In particular, if the representative agent’s uncertainty attitude is persistent and uncertainty is transitory, then the future realizations of the increasing part of the pricing kernel projected onto returns should be significantly positively correlated with current market uncertainty at short horizons (which is when the signal of the uncertainty process will be strongest), and significantly positively correlated with current market optimism at long horizons (as the shock to uncertainty will have largely dissipated at longer horizons, in which case the persistent optimism signal becomes easier to detect). We test this prediction with our proxies for market optimism and market uncertainty and find strong empirical support.

To elaborate on our third contribution, Yu and Yuan (2011) and Brenner and Izhakian (2018) show, respectively, that market sentiment and market uncertainty help to identify the risk-return tradeoff for the aggregate market. Our approach unifies these two findings theoretically by predicting how market volatility, market uncertainty, and market optimism interact in a closed-form representation of the equity premium to explain time variation in the risk-return tradeoff. We

Contemporary news articles also report a large wave of young individual investors rushing to invest in technology stocks (<https://www.cnet.com/tech/tech-industry/online-investors-come-of-age-at-25/>).

test this prediction with empirical measures of market volatility, market uncertainty, and market optimism, and find strong empirical support.

Collectively, the main contributions link equilibrium asset pricing theory and behavioral factors (market optimism and Knightian uncertainty) to provide an explanation for four classes of puzzles facing the traditional CCAPM (puzzles pertaining to matching basic stock market moments, predictability puzzles, the pricing kernel puzzle, and identification of the market risk-return tradeoff). The three contributions add to the literature on the α -maxmin multiple priors model in an asset pricing context (Zimper, 2012), by providing the first empirical study of this asset pricing model. They also extend the empirical literature on the α -maxmin multiple priors model from individual choice anomalies that challenge EU (*e.g.*, Baillon and Bleichrodt, 2015; Dimmock et al., 2015; Kocher et al., 2018), to basic puzzles for the aggregate stock market that pose challenges for an EU representative agent. Our approach thus provides a further unification between individual choice anomalies and market anomalies.¹¹

In addition to the main contributions, we provide three secondary contributions. First, we develop the NEO-EU asset pricing model from Zimper (2012) into a tractable empirical framework, where all quantities of interest are solved in closed form as functions of the state variables. This framework is potentially useful as an extension to the classical CCAPM since it incorporates a role for disaster risk, positive skewness, Knightian uncertainty, and market sentiment, thereby linking four broad strands of the asset pricing literature. Second, the framework highlights the role of the conditional market volatility, market uncertainty, and market optimism as three dynamic state variables that are subject to shocks. To the best of our knowledge, these state variables have not previously been unified in a consumption-based asset pricing framework. Despite this, they constitute three common approaches for summarizing the state of the market. Third, we derive predictions for how standard market variables vary with changes in market uncertainty and market optimism that we test with standard empirical proxies for market uncertainty and market sentiment, respectively, and for which we find strong support. Our approach thereby provides a connection between equilibrium asset pricing theory and the large empirical literature on proxies for market uncertainty and market sentiment.

The paper is organized as follows: Section 2 presents the model. Section 3 calibrates the model to match the risk-free rate, equity premium, variance risk premium, and risk-neutral entropy. Section 4 applies the model to develop and test a new mechanism for return predictability that can explain and unify predictability puzzles. Section 5 applies the model to address the pricing kernel puzzle. Section 6 applies the model to identify the risk-return tradeoff for the aggregate market. Section 7 discusses two broader themes: the interpretation of market optimism, market uncertainty, and market volatility as state variables, and the differences between the present approach and approaches based on rare events (Tsai and Wachter, 2016) or good and bad uncertainties (Bekaert and Engstrom, 2017; Segal et al., 2015). Section 8 concludes.

¹¹In the Online Appendix, we show that using the values of the representative agent's preferences from our calibration is sufficient to explain three major empirical violations of EU in laboratory experiments (the Allais paradox, the common ratio effect, and the fourfold pattern of risk preferences).

2. Model

2.1. Representative Agent

The representative agent we study maximizes a weighted average of the expected utility of an uncertain prospect and the Hurwicz value of the prospect (Chateauneuf et al., 2007) which takes a convex combination of the best and worst-case utilities.

Let there be a set S of possible states of nature, a set $\mathcal{C}$ of consumption levels, and a set $\mathcal{F}$ of acts where an act, $f : S \rightarrow \mathcal{C}$ assigns a consumption level to each state.

DEFINITION 1: A NEO-EU agent has the following value function for an act f :

$$\tilde{E}(f) = (1 - \gamma)E[u(C)] + \gamma[\alpha u(\overline{C}) + (1 - \alpha)u(\underline{C})] \quad (1)$$

In (1), $E[u(C)]$ is the agent's expected utility from consumption under act f with respect to her subjective probability distribution, while $u(\overline{C})$ and $u(\underline{C})$ are, respectively, the utility from the best-case and worst-case consumption levels across states under f . These preferences separate the agent's beliefs, her uncertainty attitude, α , ranging from pure uncertainty aversion or pure pessimism ($\alpha = 0$) to pure uncertainty seeking or pure optimism ($\alpha = 1$), and her perceived level of Knightian uncertainty, γ , ranging from no Knightian uncertainty ($\gamma = 0$) to complete uncertainty ($\gamma = 1$). As the NEO-EU model nests rational expectations ($\gamma = 0$), and pure uncertainty aversion ($\gamma \in (0, 1]$, $\alpha = 0$), the agent we study is a natural choice for testing whether or how far asset prices imply deviation from two main theoretical benchmarks in the literature.

Wakker (2010) notes that the probability weighting function embedded in (1), is among the most promising families of weighting functions in the literature and "the interpretation of its parameters is clearer and more convincing than with other families". By overweighting the extreme outcomes, the NEO-EU agent can explain the behavior of a consumer who purchases both lottery tickets and insurance, which has been a challenge for EU since its inception (Friedman and Savage, 1948).

2.2. General Equilibrium

We consider a Lucas-tree economy with a NEO-EU representative agent. The economy has one risky asset representing the aggregate stock market. The economy has a risk-free bond of net zero supply. The infinitely-lived NEO-EU representative agent faces an optimization problem every period. At time t , the agent chooses its level of consumption and investment to maximize

$$u(C_t) + E_t \sum_{j=0}^{\infty} \beta^j \left((1 - \gamma_{t+j})u(C_{t+1+j}) + \gamma_{t+j}[\alpha_{t+j}u(\overline{C}_{t+1+j}) + (1 - \alpha_{t+j})u(\underline{C}_{t+1+j})] \right), \quad (2)$$

where $E_t u(C_{t+1})$ is the time t expected utility of consumption in period $t + 1$, and $u(\overline{C}_{t+1})$ and $u(\underline{C}_{t+1})$ are utilities from the perceived best (optimistic) and worst (pessimistic) case consumption levels in period $t + 1$. Note that $E_t u(C_{t+1})$, $u(\overline{C}_{t+1})$ and $u(\underline{C}_{t+1})$ are all time t information. γ_t and α_t represent the agent's perceived *uncertainty* and *uncertainty attitude* at time t . The budget

constraint for each period t is

$$C_t + S_t P_t + B_t = S_{t-1}(D_t + P_t) + B_{t-1} R_{f,t-1}, \quad t \geq 0.$$

S_t and B_t represent stock and bond holdings, whose gross returns between t and $t+1$ are $R_{t+1} = \frac{D_{t+1} + P_{t+1}}{P_t}$ and $R_{f,t}$, respectively. We let α_t and γ_t be exogenous state variables that are subject to shocks and the agent takes the dynamics of α_t and γ_t into account when optimizing.

A classic result by Dow and da Costa Werlang (1992) implies that there is a range of prices at which a NEO-EU agent will neither buy nor short-sell an asset. The buying and short-selling prices need not be identical for such an agent. As a simple and practical way to resolve this indeterminacy, throughout, we assume that the prevailing market price is the buying price.¹² One can interpret this assumption as assuming there is a market maker who always sets the price at the buying price and this is public information. Consequently, every trader that buys the stock knows she can only sell it at the buying price. This property, along with the condition that the agent takes the dynamics of α_t and γ_t into account when optimizing ensures that the setup is dynamically time consistent (similar to models with shocks to utility parameters).

The agent's optimization yields the Euler equation

$$(1 - \gamma_t) E_t M_{t+1} R_{t+1} + \gamma_t (\alpha_t \bar{M}_{t+1} \bar{R}_{t+1} + (1 - \alpha_t) \underline{M}_{t+1} \underline{R}_{t+1}) = 1, \quad (3)$$

where $M_{t+1} := \beta \frac{u'(C_{t+1})}{u'(C_t)}$, and $\bar{M}_{t+1}$ and $\bar{R}_{t+1}$ ($\underline{M}_{t+1}$ and $\underline{R}_{t+1}$) are associated with the next period's optimistic (pessimistic) outcomes. The Euler equation for the gross risk-free return $R_{f,t}$ is

$$(1 - \gamma_t) E_t M_{t+1} R_{f,t} + \gamma_t (\alpha_t \bar{M}_{t+1} R_{f,t} + (1 - \alpha_t) \underline{M}_{t+1} R_{f,t}) = 1. \quad (4)$$

Subtracting (4) from (3) and rearranging terms yields the equity premium:

$$E_t R_{t+1} - R_{f,t} = \underbrace{-\frac{Cov_t(M_{t+1}, R_{t+1})}{E_t [M_{t+1}]}}_{\text{Risk Premium}} + \underbrace{\frac{[(R_{f,t} - \bar{R}_{t+1}) \bar{M}_{t+1}] \alpha_t \gamma_t}{E_t [M_{t+1}] (1 - \gamma_t)}}_{\text{Speculation Premium}} + \underbrace{\frac{[(R_{f,t} - \underline{R}_{t+1}) \underline{M}_{t+1}] (1 - \alpha_t) \gamma_t}{E_t [M_{t+1}] (1 - \gamma_t)}}_{\text{Uncertainty Premium}}. \quad (5)$$

Equation (5) decomposes the equity premium into three terms. The first term is the standard CCAPM risk premium reflecting covariance of asset returns with M_{t+1} . The second term is a discount associated with the lottery demand for skewness. We refer to it as a *speculation premium* and it is negative, reflecting that investors pay to hold stocks that are more exposed to market optimism (or a market boom). The speculation discount becomes larger in magnitude with an increase in market optimism, α_t , with an increase in market positive skewness, $\bar{R}_{t+1}$, or with an increase in market uncertainty, γ_t . The third term is an *uncertainty premium* which becomes larger in magnitude with an increase in market uncertainty aversion, $(1 - \alpha_t)$, with an increase in market disaster risk (lower $\underline{R}_{t+1}$), or with an increase in market uncertainty, γ_t .

The equity premium in Equation (5) predicts that the realized speculation premium is strongest

¹²This assumption is consistent with the literature (cf. Epstein and Wang (1994), Section 3 and Footnote 10).

following periods when market optimism (α_t) is high and that the realized uncertainty premium is strongest following periods when α_t is low. Consistent with this prediction, Kerestecioglu et al. (2021) provide empirical evidence for the presence of both a positive uncertainty premium and a negative ‘mispricing’ premium and find that the “mispricing effect is only observed following a period of high sentiment,” whereas the uncertainty-premium effect is only observed following a low-sentiment period. In evaluating their data, they conclude “This evidence could be a starting point for further theoretical development of a unified risk premium and mispricing model.” One might interpret Equation (5) as a step in this direction.

Representation (5) generalizes the Consumption CAPM to include a role for market optimism (α_t), Knightian uncertainty (γ_t), positive skewness ($\bar{R}_{t+1}$), and disaster risk ($\underline{R}_{t+1}$), thereby linking four primary strands of the asset pricing literature. Since (5) generalizes the CCAPM based on an EU representative agent to permit deviations from rational expectations consistent with the NEO-EU model, we refer to (5) as the *NEO-EU CAPM*.

2.3. Price-Dividend Ratio and Return

We explore the effect of optimism and uncertainty on the price-dividend ratio, which we use in our discussion of return predictability in Section 4. To illustrate the mechanism in its simplest form, we put a simple structure on the consumption and dividend growth. Specifically, we assume $\log\left(\frac{C_{t+1}}{C_t}\right) = g + \sigma z_{t+1}$, and $\log\left(\frac{D_{t+1}}{D_t}\right) = g + \phi\sigma z_{t+1}$, where z_t is an *iid* standard normal random variable, and $\phi > 1$ captures leverage similar to Abel (1999).

We operationalize the concept of the best and worst case scenarios for a NEO-EU agent by assuming that the agent truncates the tails of a normal distribution:

$$X \sim \mathcal{N}(\mu, \sigma) \Rightarrow \begin{cases} \bar{X} = \mu + \bar{\xi}\sigma \\ \underline{X} = \mu - \underline{\xi}\sigma \end{cases},$$

where parameters $\bar{\xi}$ and $\underline{\xi}$ are large enough that truncation virtually leaves the distribution unchanged. This behavior is consistent with Camerer and Kunreuther (1989) who cite evidence that consumers truncate the left tail in insurance purchases, and with Smith (2014) who argues that truncating the right tail explains the St. Petersburg paradox.

Since $R_{t+1} = \frac{PD_{t+1}+1}{PD_t} \frac{D_{t+1}}{D_t}$, log-linearizing the Euler equation (3) and finding the forward looking solution for $pd_t := \log(PD_t)$ as in Campbell and Cochrane (1999) yields

$$pd_t = \psi_0 + \psi_1 h_t, \quad \text{where } h_t := \gamma_t((\bar{\xi} + \underline{\xi})\alpha_t - \underline{\xi}). \quad (6)$$

where ψ_0 and ψ_1 are positive constants that depend on model deep parameters and h_t represents the contribution of the Hurwicz part in the NEO-EU optimization. Appendix C provides the details. From (6), optimism α_t increases prices, and importantly, uncertainty γ_t decreases prices if and only if α_t is low (*i.e.*, $\alpha_t < \frac{\underline{\xi}}{\bar{\xi} + \underline{\xi}}$, which holds on average in our calibration).

Using the approximation for the log price-dividend ratio in (6), we can approximate the log-

returns r_{t+1} , which helps to better understand the mechanisms that yield return-predictability. Log-linearizing $R_{t+1} = \frac{PD_{t+1}+1}{PD_t} \frac{D_{t+1}}{D_t}$, we have

$$r_{t+1} = r_0 + \omega pd_{t+1} - pd_t + \sigma \phi z_{t+1}, \quad (7)$$

where $\omega \equiv PD/(PD+1) < 1$, $r_0 \equiv \log(PD+1) - \omega pd + \omega + g$, and PD is the unconditional mean of PD_t . Since dividend growth is assumed *iid*, the predictable part of the return resides in pd_{t+1} . Thus, from (6), the return predictability hinges on the persistence of the processes of α_t and γ_t .

3. The Equity Premium, VRP, and VIX-SVIX Puzzles

The NEO-EU representative agent adds two parameters $\bar{\xi}$ and $\underline{\xi}$, and two state variables α_t and γ_t to the Consumption CAPM with temporal discount rate $\beta = e^{-\rho}$ and elasticity of intertemporal substitution θ^{-1} . We set $\theta = 1$ (log utility), which is in the range consistent with laboratory estimates of the CRRA utility parameter (Holt and Laury, 2002), and allows us to use the nominal market returns, risk-free rate, and consumption growth (since with $\theta = 1$, inflation drops out of our Euler equations).

We use $\bar{\xi}$ to match the top 99.95% percentile of the monthly returns in CRSP. Then, we simultaneously match four moments: (i) risk-free rate R_f ; (ii) equity premium EP; (iii) risk-neutral entropy VIX²; (iv) risk-neutral variance of the excess market return SVIX², to find the unconditional means of α and γ together with parameters $\underline{\xi}$ and ρ . Our data consists of approximately thirty years of monthly observations from January, 1990 (the first year of the VIX index) through December, 2018.^{13,14} By matching VIX and SVIX, we match the variance risk premium and the difference VIX-SVIX, each of which the current literature finds hard to match (Martin, 2017; Bekaert et al., 2020).¹⁵ All data sources are listed in the Online Appendix.

3.1. Model Calibration

If $E_t h_{t+1}$ is linear in h_t , Equations (6) and (7) imply that $E_t r_{t+1} = \theta_0 + \theta_1 pd_t$, where θ_0 and θ_1 are constants. Then, the conditional normality of the error term implies

$$\begin{bmatrix} \log(R_{t+1}) \\ \log(\frac{C_{t+1}}{C_t}) \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} x_t \\ \mu \end{bmatrix}, \begin{bmatrix} q_t^2 & q_t \sigma \eta \\ q_t \sigma \eta & \sigma^2 \end{bmatrix} \right), \quad (8)$$

¹³The only exceptions are for the option-based measures SVIX and the market crash probability, CP, (reported in Martin, 2017, SVIX and CP data covers the period January, 1996, through January, 2012), the option implied correlation, IC (Buss et al., 2019) which covers the period January, 1996 through December, 2017, and the right tail of the pricing kernel, $\tilde{M}$ (Sichert, 2020) which covers the period February, 1996 through December, 2018.

¹⁴Matching the four moments does not hinge on the specific percentile used for the upper bound on returns.

¹⁵Using a non-expected utility model based on disappointment aversion Schreindorfer (2020) comes closer to matching the VIX-SVIX difference than previous macro-finance models, although the difference generated by his calibration (0.44%) remains smaller than the 0.77% difference reported in Martin (2017).

where $x_t := \theta_0 + \theta_1 pd_t$. Since we have not specified dynamics of α_t and γ_t , we can directly put a linearity assumption on $E_t h_{t+1}$, with the understanding that the error term can in general have complex dynamics (or simply follow a GARCH process). Given (8), the risk-free rate and equity premium (without log-linearization) are given by (11) and (12) in the Appendix. From Martin (2017), VIX from t to $t+1$ satisfies $VIX_t^2 = 2(\log \tilde{E}_t^Q R_{t+1} - \tilde{E}_t^Q \log R_{t+1})$ and SVIX from time t to $t+1$ satisfies $SVIX_t^2 = \text{Var}_t^Q(\frac{R_{t+1}}{R_{f,t}})$, where Q denotes the risk-neutral measure. The expressions for the risk-neutral entropy and risk-neutral variance of the excess market return are given by (13)-(14) in the Appendix.

To find the conditional expected return (x_t) and conditional standard deviation of returns (q_t), we estimate a parsimonious GARCH(1,1) model. That is, we assume the error term in $r_t = \theta_0 + \theta_1 pd_{t-1} + \epsilon_t$ follows a GARCH(1,1) structure. Table OA.5 in Online Appendix shows that pd significantly predicts the return up to twelve months ahead, but the coefficient is most significant at one month (which is our model). Both Akaike and Bayesian information criteria pick the GARCH(1,1) model over models with more ARCH and GARCH terms.

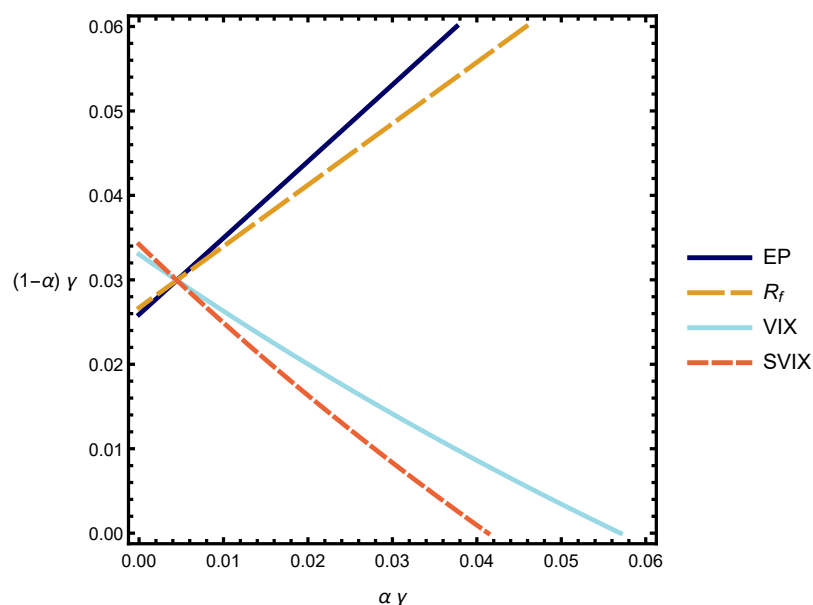
We replace μ and σ with the unconditional mean and standard deviation of monthly log consumption growth (0.30% and 0.32%). The correlation between monthly log consumption growth and monthly log returns, η , is 17%. The mean of x_t and q_t from the GARCH regression are respectively, 0.89% and 4.10%. Figure 1 summarizes these monthly averages.

We set $\bar{\xi}$ such that $\overline{R_{t+1}}$ matches the return corresponding to 99.95% of the empirical cumulative distribution of monthly returns from CRSP, ($\overline{R_{t+1}} = 20.50\%$). Given the average monthly market volatility of 4.10%, we have $\bar{\xi} = 20.50/4.10 = 5.00$. Substituting the monthly averages for μ , σ , x_t , q_t , ρ , and η , and setting $\bar{\xi} = 5.00$, and $\theta = 1$, we simultaneously solve (11)-(14) to find $\rho = 0.01\%$, $\underline{\xi} = 6.60$, $\alpha = 0.138$, and $\gamma = 0.036$, tabulated below Figure 1. Figure 1 shows that in the space of $\gamma(1 - \alpha)$ and $\gamma\alpha$ (that span h_t), and after substituting all parameters and monthly averages, the loci of the points that satisfy the four equations (11)-(14) have a unique intersection.

Brief discussion of matched moments. ρ serves to match the risk-free rate and has a small effect on SVIX. Effectively, $\underline{\xi} = 6.60$ serves to match the average difference between VIX and SVIX. Martin (2017) shows that VIX is more sensitive to the left tail of the risk-neutral distribution and in comparison, SVIX is more sensitive to the right tail. Thus, the VIX-SVIX difference provides information about the heaviness of the left tail relative to the right tail. Our calibration of $\underline{\xi}$ and $\bar{\xi}$ first fixes the right tail weight, $\bar{\xi}$, and then uses the VIX-SVIX difference to determine the value of the left tail weight $\underline{\xi}$, enabling us to match the difference. Note that $VRP_t = R_{f,t} SVIX_t^2 - q_t^2$. Consequently, matching SVIX and R_f means that our calibration also matches VRP.

We find $\underline{\xi} = 6.60$, which implies $\underline{R_{t+1}} = -27.06\%$. Approximately 0.05% of the empirical distribution of returns in our sample falls below this value. The truncated distribution with $\bar{\xi} = 5$ and $\underline{\xi} = 6.60$ contains almost exactly 99.9% of the empirical distribution. We could equivalently match the lower and upper side of the empirical return distribution ($\bar{R}$ and $\underline{R}$) at 0.05% and 99.95%, and observe that the same calibration matches the VIX-SVIX and the VRP moments.

Figure 1. Top panel: Loci of points in the $(\gamma(1-\alpha), \gamma\alpha)$ space that match equity premium (EP), risk-free rate (R_f), risk-neutral entropy (VIX), and risk-neutral variance of excess return (SVIX). Bottom panel: Monthly averages from data and matched annualized unconditional moments.



Notes: μ and σ are the mean and standard deviation of consumption growth; η is the correlation between consumption growth and the market return; x and q are the conditional expected market return and volatility; model parameters are intertemporal elasticity of substitution, θ^{-1} , temporal discount rate, $e^{-\rho}$, and $\bar{\xi}$ and $\underline{\xi}$; R_f is the risk free rate, and EP is the equity premium. SVIX and VIX are from Martin (2017), and VRP is calculated as $R_f \text{SVIX}^2 - 12q^2$.

As benchmark values for α and γ , note that $\gamma = 0$ corresponds to EU. Second, if $\gamma > 0$, the focus on only negative outcomes implied by the uncertainty aversion and robust control literatures suggests the benchmark $\alpha = 0$. Our calibration yields the unconditional values $\alpha = 0.138$ and $\gamma = 0.036$. Hence, the model unconditionally matches the key moments with a weight of above 0.96 on the expected utility component of the agent's preferences. The deviation from EU is systematic with a primary bias toward uncertainty aversion ($\gamma(1 - \alpha) = 0.031$) and a small amount of optimism ($\gamma\alpha = 0.005$).

Elasticity of equity premium with respect to state variables. It is instructive to know how changes in state variables (on average) affect the equity premium. In particular, we look at the following counterfactual exercise. We assume that the system is at its unconditional mean, and we change a particular state variable by one percentage point while keeping everything else fixed, and report how the equity premium changes. We observe that raising α from 0.138 to 0.148 lowers the annual equity premium by 0.21 percentage points. Raising γ from 0.036 to 0.046 raises the annual equity premium by 2.30 percentage points, and raising q from 0.041 to 0.051 raises the equity premium by 1.58 percentage points. Thus, as one would expect, unconditionally, small changes in uncertainty have a larger effect compared to small changes in uncertainty attitude.

Robustness check with respect to tail parameters. We emphasize that the success of our calibration exercise above does not hinge on having large tail cut-off parameters $\underline{\xi}$ and $\bar{\xi}$. For instance, we could choose the tail parameters as small as two standard deviations from the mean (*i.e.*, $\underline{\xi} = \bar{\xi} = 2$), and then calibrate the three remaining parameters α , γ , and ρ to match the annual values of the equity premium (0.077), variance risk premium (0.025), and risk-free rate (0.027). This calibration yields $\gamma = 0.40$ and $\alpha = 0.39$ (and a small positive ρ as usual), which is very close to Dimmock et al. (2015)'s estimate of $\gamma = 0.40$ and $\alpha = 0.44$ in an experiment with a representative sample of the U.S. population. This alternative calibration shows that the present framework can simultaneously match the risk-free rate, equity premium, and variance risk premium without assuming asymmetric or extreme tail parameters, or extreme risk aversion, or extreme uncertainty aversion, and with values of α and γ close to experimental estimates.

Note that choosing two standard deviations is close to assuming that the investor (unconditionally) has behavioral cut-off parameters associated with a "market correction" (that is, almost a ten percent change in market prices, given that $q = 0.041$). Of course, market corrections happen much more frequently compared to the type of rare events associated with $\underline{\xi} = 6.6$ and $\bar{\xi} = 5$.

4. Return Predictability Puzzles

We impose only a mild assumption on the dynamics of α_t and γ_t : shocks to α_t are typically large and persistent, while shocks to γ_t are mostly transitory (due to resolution of uncertainty).¹⁶

¹⁶Recent empirical studies suggest that utility parameters controlling uncertainty perception (γ_t) and uncertainty attitude (α_t) are time dependent (Grevnbrock et al., 2020).

To the extent that α_t represents market optimism, its natural empirical proxy is the Baker and Wurgler (2006) sentiment index (BW), which is widely used to measure market optimism. One might note that α_t pertains specifically to optimism about returns at the right tail of the return distribution. Consistent with this property, the BW index is also related to optimism toward the right tail. We provide evidence of this property in Section 5 by showing that BW predicts variation in the increasing portion of the stochastic discount factor at the right tail of the return distribution. Further evidence of the link between BW and the right tail is provided by Fong and Toh (2014) who show that BW strongly predicts time variation in the returns on lottery-like stocks.

To the extent γ_t represents Knightian uncertainty, a natural empirical proxy is the VRP which has been used as a measure of uncertainty that is separate from risk by Bali and Zhou (2016). As Zhou (2018) comments: “the difference between option-implied and expected (or realized) variances effectively removes the volatility level or risk component, and the resulting variance risk premium is indeed a pure measure of the uncertainty premium, or of economic uncertainty itself”. Consistent with our assumption that α_t (γ_t) is a persistent (transitory) process, we find for our sample that BW (VRP) has a monthly autocorrelation of 0.97 (0.55).

The dynamics of α_t and γ_t generate a new mechanism for return predictability which provides a unified explanation for three sources of return predictability: the variance risk premium (Zhou, 2018), market crash risk (Martin, 2017), and market correlation (Buss et al., 2019). Importantly, this mechanism makes the new prediction that the positive return predictability of these variables comes from periods in which PD is low. The mechanism further predicts positive return predictability of the U-shape of the stochastic discount factor arising from periods in which PD is low. We test each of these predictions and find strong empirical support. The same mechanism generates the standard negative predictability of PD which we discuss first.

Consistent with our calibration, we assume that $\underline{\xi}$ and $\bar{\xi}$ are large enough that the probability of more extreme observations for a standard normal variable is virtually zero, and also $\bar{\xi} < \underline{\xi}$.

4.1. Return Predictability of the Price-Dividend Ratio

For a persistent mean-reverting optimism process, α_t , high optimism today (high α_t) means, on average, lower future optimism (α_{t+1}). Thus, $pd_{t+1} < pd_t$, on average, when α_t is high. From (7), a declining PD implies that future returns also decline. The more persistent the process for α_t , the longer it takes for the declining PD to clearly manifest itself, and hence, we see a larger drop (reverting back to the unconditional mean) over longer horizons.

PREDICTION 1: *PD negatively predicts future market returns. Moreover, for a more persistent process of α_t , the predictability manifests itself better over longer horizons.*

Campbell and Shiller (1988) found that PD predicts the market return at long horizons.

4.2. Return Predictability of the Variance Risk Premium

The VRP positively predicts market returns at short horizons (Bollerslev et al., 2009). To motivate a new mechanism for explaining return predictability, we observe that the NEO-EU representative agent implies the following prediction for how market optimism α_t affects PD and how market uncertainty γ_t affects the VRP.

PREDICTION 2: *PD is increasing in optimism. VRP is increasing in uncertainty.*

The first part of Prediction 2 can be already seen in Equation 6. We next explain the second part of the prediction. Recall that $\text{VRP}_t \equiv \frac{1}{R_{f,t}} \text{Var}_t^Q R_{t+1} - \text{Var}_t^P R_{t+1}$ (Martin, 2017), where Var_t^Q and Var_t^P are the conditional variance under the risk-neutral and physical measures. Recall $\tilde{E}X \equiv (1 - \gamma)EX + \gamma(\alpha\bar{X} + (1 - \alpha)\underline{X})$, and hence,

$$\tilde{\text{Var}}X = (1 - \gamma)E(X - \tilde{E}X)^2 + \gamma[\alpha(\bar{X} - \tilde{E}X)^2 + (1 - \alpha)(\underline{X} - \tilde{E}X)^2]. \quad (9)$$

Thus, an increase in γ reduces the weight on the middle values of the X distribution and puts that weight on the extreme values $\bar{X}$ and $\underline{X}$ (that depend on $\bar{\xi}$ and $\underline{\xi}$), increasing the variance $\tilde{\text{Var}}X$ (and hence, the VRP). Moreover, an increase in α only shifts the weight from the low extreme value to the high extreme value, which will reduce $\tilde{\text{Var}}X$ when $\bar{\xi} < \underline{\xi}$.

When X is a return, under the risk-neutral measure, $\tilde{E}X$ equals the risk-free rate, whose response to a change in γ is small and ambiguous. However, extreme values $\bar{X}$ and $\underline{X}$ directly depend on the exogenous parameters $\bar{\xi}$ and $\underline{\xi}$. Sufficiently large $\bar{\xi}$ and $\underline{\xi}$ guarantee that variation in $\tilde{\text{Var}}X$ (and hence, in VRP) is primarily due to variation in γ . Consequently, a low PD and high VRP on average mean low optimism and high uncertainty. This means that in the near future, optimism is still low (due to its high persistence) but the high uncertainty is mostly resolved (due to its low persistence). As uncertainty resolves in the near future (γ decreases), the future PD and hence, the future return increases, leading to Prediction 3:

PREDICTION 3: *The VRP positively predicts market returns when PD is low.*

When PD is high, on average α is high, which makes the effect of γ (and hence, VRP) on PD disappear for α close to $\frac{\underline{\xi}}{\bar{\xi} + \underline{\xi}}$. For $\alpha > \frac{\underline{\xi}}{\bar{\xi} + \underline{\xi}}$, the effect is theoretically reversed. However, $\frac{\underline{\xi}}{\bar{\xi} + \underline{\xi}}$ is much larger than the unconditional mean of α and may rarely occur in the data. For $\alpha < \frac{\underline{\xi}}{\bar{\xi} + \underline{\xi}}$, the effect of γ on pd in (6) is largest when α is close to zero (associated with low PD).

The dynamics of α and γ generate short-run predictability of VRP and long-run predictability of PD, and these sources of predictability are complementary, due to the persistent nature of α which drives long-run price variation and the transitory nature of γ which drives short-run price variation. As Miao et al. (2019) note: “the short-run and long-run predictability seem to be complementary in the sense that when dividend yield or P/E ratio is included in the regressions, the predictability of the variance premium is not crowded out but often is enhanced. It is not clear that any existing consumption-based asset pricing model can replicate such a puzzling phenomenon.” In the present

framework, this complementary predictability arises from the main assumption of our predictability mechanism: the high (low) persistence of α (γ).

4.3. Return Predictability of Market Tail Risk

Market tail risk such as the risk of a major stock market crash also predicts future returns as can be demonstrated from the option-implied crash probability (CP) in Martin (2017). As a simple measure of tail risk, define the *risk-neutral tail risk* as the risk-neutral probability that $C_{t+1} = \underline{C}_{t+1}$. For the NEO-EU agent, risk-neutral tail risk is proportional to $\gamma_t(1 - \alpha_t)$.

PREDICTION 4: *The risk-neutral tail risk is increasing in uncertainty and decreasing in optimism.*

Consequently, a low PD and high tail risk on average means low optimism and high uncertainty. The NEO-EU representative agent model then implies return predictability for the risk-neutral tail risk through a mechanism analogous to that explained for VRP return predictability.

PREDICTION 5: *Risk-neutral tail risk positively predicts market returns when PD is low.*

4.4. Return Predictability of Market Correlation

Variation in the market-implied correlation between stocks also positively predicts market returns (Buss et al., 2019). Under the NEO-EU representative agent model, the correlation between asset returns depends on the shocks to state variables α_t and γ_t . To see this, consider the following simple process for individual asset i 's dividend growth

$$\log \left(\frac{D_{i,t+1}}{D_{i,t}} \right) = g + \phi \sigma \eta_i z_{t+1} + \epsilon_{i,t+1}, \quad z_{t+1} \perp \epsilon_{i,t+1}.$$

In this setting, the “idiosyncratic” part $\epsilon_{i,t+1}$ is assumed orthogonal to z_{t+1} (and hence, to consumption growth), and thus does not affect asset i 's return in the best and worst-case scenarios of consumption growth (a simplifying assumption). The price-dividend ratio of asset i is

$$pd_{i,t} = \psi_0 + \psi_1 \eta_i h_t, \quad \text{where } h_t = \gamma_t((\underline{\xi} + \bar{\xi})\alpha_t - \underline{\xi}).$$

Asset prices then co-move more with changes in market optimism (*i.e.*, correlation across assets is higher due to a change in α) when market uncertainty is high. Asset prices co-move more with changes in market uncertainty (*i.e.*, correlation across assets is higher due to a change in γ) when market optimism is low (given $\alpha < \frac{\bar{\xi}}{\underline{\xi} + \bar{\xi}}$).

PREDICTION 6: *Market correlation is increasing in uncertainty and decreasing in optimism.*

Consequently, a low PD and high correlation between assets on average means low optimism and high uncertainty. Similar to the mechanism in Predictions 3 and 5, the NEO-EU representative agent model implies positive return predictability of market correlation when PD is low.

PREDICTION 7: *Market correlation positively predicts market returns when PD is low.*

4.5. Return Predictability of the Pricing Kernel U-Shape

As we discuss in Section 5, the stochastic discount factor or pricing kernel of our representative agent has a U-shape, increasing at the right tail of the return distribution as documented empirically (Bakshi et al., 2010; Sichert, 2020). In the present framework, the increasing part of the pricing kernel is increasing in the product $\alpha_t \gamma_t$.

In our framework, variation in PD is primarily driven by variation in α_t , as both are highly persistent. In contrast, γ_t only contributes to short-run fluctuations of PD around its long-run trend. Recall from Section 2 that uncertainty γ_t decreases PD if and only if α_t is low. A low PD and pronounced pricing kernel U-shape (high $\alpha_t \gamma_t$) means low optimism (which suppresses PD) and high uncertainty (which suppresses PD and increases $\alpha_t \gamma_t$). This means that in the near future, optimism is still low (due to its high persistence) but the high uncertainty is mostly resolved (due to its low persistence). As uncertainty resolves in the near future (γ_t decreases), the future PD and hence, the future return increases.

We then have the following prediction about the increasing part of the pricing kernel.

PREDICTION 8: *Time variation in the U-shape of the pricing kernel predicts market returns: The increasing part of the U-shape positively predicts market returns when PD is low.*

To our knowledge, the pricing kernel U-shape has not been previously applied to predict market returns. Consequently, we go beyond providing a mechanism for explaining the return predictability of existing predictors and identify a new candidate return predictor as summarized in prediction 8.

4.6. Return Predictors, Market Optimism, and Market Uncertainty

Predictions 4 and 6 predict that market crash risk and market correlation are each increasing in market uncertainty and decreasing in market optimism. We test these predictions using established proxies for market uncertainty and market optimism. The results are shown in Table 1 using the market crash probability over a one month horizon (CP) from Martin (2017), and the option-implied correlation over a one month horizon (IC) from Buss et al. (2019) as dependent variables. As proxies for market optimism and market uncertainty, we use the BW index of market sentiment and the VRP, respectively. Table 1 displays correlations between BW and PD, CP, and IC, as well as the lag 1 autocorrelation for BW. As predicted, PD is increasing in optimism, while CP and IC are decreasing in optimism. The second row in Table 1 displays correlations between VRP and PD, CP, and IC, as well as the lag 1 autocorrelation for VRP. We find that PD is decreasing in uncertainty while CP and IC are increasing in uncertainty, as predicted by the model. Further, optimism explains a larger portion of the variation in PD, whereas uncertainty explains a larger portion of the variation in CP and IC. Table OA.1 in the online appendix evaluates Predictions 4 and 6 using a broader set of empirical optimism and uncertainty proxies.¹⁷

¹⁷Table OA.1 uses three proxies for market optimism: the Baker and Wurgler (2006) market sentiment index (BW), the Huang et al. (2015) market sentiment index (HJTZ), and the Michigan index of consumer sentiment (ICS). The table also uses three uncertainty proxies: VRP, VRPx (a VRP measure constructed from the VIX index of the Chicago Board of Options Exchange), and the economic policy uncertainty (EPU) index of Baker et al. (2016).

Table OA.1 serves as a robustness check, demonstrating that the predicted relationships between optimism, uncertainty, crash risk, and market correlation hold more broadly. The table also provides, to our knowledge, the first regressions of crash risk and market correlation that simultaneously control for variation in all three model state variables (optimism, uncertainty, and market volatility). The table reveals that these three state variables explain up to 83% of the variation in the market crash probability and up to 51% of the variation in market correlation.

Table 1. Correlations of Return Predictors with Proxies for Optimism and Uncertainty.

	PD	CP	IC	AC(1)
BW	0.50***	-0.16**	-0.38***	0.97***
VRP	-0.22***	0.66***	0.53***	0.55***

Notes: The table displays correlations between the Baker and Wurgler (2006) market sentiment index (BW) (a proxy for market optimism) and three return predictors (PD, the market crash probability (CP) of Martin (2017), and the implied correlation (IC) from Buss et al. (2019), as well as the lag-one auto-correlation (AC) for BW. The second row displays the corresponding correlations for VRP. All series are monthly. The correlations span the VRP sample period from January, 1996 through January, 2012. (** $p < 0.05$, *** $p < 0.01$).

4.7. Empirical Return Predictability when PD is Low

Although PD, VRP, CP, IC, and the increasing part of the pricing kernel are quite different quantities, our framework provides a common mechanism underlying their return predictability as arising from the interactions of a persistent process α_t and a transitory process γ_t . In this section, we test model predictions 3, 5, 7 that return predictability of VRP, CP, and IC comes from periods in which PD is low. For each return predictor (VRP, CP, IC), we regress the market excess return against that predictor and that predictor conditional on periods in the lowest decile of PD values, both with four and five-month lags. Table 2 shows the results for the VRP, constructed from SVIX, and for a common proxy for the VRP constructed from VIX. Table 3 shows the results for CP and IC. Table 4 shows the analogous results for the increasing part of the pricing kernel.

Conditioning on low PD periods uncovers predictability that is not observed unconditionally. For instance, unconditionally, our VRP measure based on Martin (2017) only has predictability at lag 5. When conditioning on low PD periods, VRP also has predictability at lags 4, 8, and 12. As the VRP is traditionally viewed as a short-horizon predictor, we focus on its predictability at the shorter horizon lags of 4 months and 5 months. To be consistent, we use the same lags (4 months and 5 months) to test for the predictability of all return predictors.

We also test Prediction 8 that time variation in the pricing kernel U-shape predicts future market excess returns. We do so by regressing the market excess return against the increasing part of the pricing kernel that generates the U-shape and that predictor conditioned on periods in the lowest decile of PD values and conditioned on periods in the highest decile of PD values at lag 4, using a measure of the increasing part of the pricing kernel from Equation (5) in Sichert (2020) which is formally defined in footnote 20 in Section 5. The measure, denoted by $\tilde{M}$ uses the

physical distribution from Bekaert and Hoerova (2014). The results are shown in Table 4. All of the predictability results remain very similar if the market excess return is replaced by the market return or the log market return.

Regressions 1-4 in the top panel of Table 2 test for predictability at a four-month lag by (1) VRP; (2) VRP and VRP conditional on low PD periods; (3) VRP and VRP conditional on NBER recessions; (4) VRP, VRP conditional on low PD periods, and VRP conditional on recessions. Regressions 5-8 are analogous but use a five-month lag. Regressions 1-8 in the bottom panel are analogous to the top panel regressions but replace VRP with VRPx. Table 3 reports analogous regressions for CP and IC. Table 4 reports analogous regressions for $\tilde{M}$.¹⁸

As shown in regressions (2) and (6) in the top panel of Table 2, return predictability of VRP comes from low PD periods. Similarly, in regressions (2) and (6) in the bottom panel of Table 2, predictability of VRPx comes from low PD periods. As shown in regressions (2) and (6) in the top panel of Table 3, return predictability of CP comes from low PD periods. As shown in regressions (2) and (6) in the bottom panel of Table 3, return predictability of IC comes from low PD periods. As shown in regressions (2) and (6) in the top panel of Table 4, return predictability of $\tilde{M}$ comes from low PD periods.

Collectively, the regressions reveal that conditioning on low PD periods subsumes the unconditional predictability when present, and uncovers conditional predictability in cases where unconditional predictability is not observed. These findings provide strong support for predictions 3, 5, 7, and 8. That the predictability in low PD periods occurs at the same time horizon for all predictors (VRP, VRPx, CP, IC, and $\tilde{M}$) further supports our claim for a common underlying mechanism.

Studies by Rapach et al. (2010), Henkel et al. (2011), Dangi and Halling (2012), Garcia (2013), and Golez and Koudijs (2018) document strong return predictability during recessions. While four of these papers include the dividend yield as one of their predictor variables, none of the previous studies used either VRP, CP, IC, or $\tilde{M}$, which are the focus of our predictability mechanism.

As a robustness check, we control for predictability due to recessions by conditioning VRP, CP, IC, and $\tilde{M}$ on the NBER recession indicator.¹⁹ We find the predictive power of VRP, CP, IC, and $\tilde{M}$ comes from low PD periods, and this predictability is distinct from and complementary to predictability during recessions. At lag 4, conditioning on low PD periods produces an R-squared of 3% for VRP, 3% for VRPx, 4.6% for CP, 4.7% for IC, and 2.9% for $\tilde{M}$. At lag 4, conditioning on both low PD periods and recessions identifies significant positive predictability from low PD periods and significant negative predictability from recessions, and the R-squared increases to 10.7% for VRP, 7.5% for VRPx, 9.9% for CP, 9.7% for IC, and 4.2% for $\tilde{M}$. The negative predictability from recessions is a curious new empirical finding that warrants further investigation in future research.

¹⁸For robustness, we include the analogous tables with OLS t-statistics in the online appendix. These results are similar to those in Tables 2, 3 and 4.

¹⁹We realize that NBER recessions are dated ex post, but that data is used for illustration purposes in previous studies of predictability in recessions.

Table 2. Return Predictability of the Variance Risk Premium in Low PD Periods

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%
L4.VRP	0.15 (0.30)	-0.17 (-0.26)	0.97*** (2.65)	1.01*** (2.83)				
L4.VRP _{PD}		1.95*** (3.09)		3.45*** (10.09)				
L4.VRP _{REC}			-1.46** (-2.07)	-2.54*** (-5.34)				
L5.VRP					0.66** (2.29)	0.39 (1.51)	0.79* (1.90)	0.81* (1.95)
L5.VRP _{PD}						1.61*** (5.78)		2.15*** (3.41)
L5.VRP _{REC}							-0.24 (-0.36)	-0.91 (-1.28)
R^2	0.001	0.030	0.032	0.107	0.018	0.038	0.019	0.048
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%
L4.VRPx	0.06 (0.17)	-0.26 (-0.54)	0.57* (1.83)	0.58* (1.88)				
L4.VRPx _{PD}		1.54*** (3.19)		2.44*** (7.67)				
L4.VRPx _{REC}			-0.85 (-1.54)	-1.72*** (-4.38)				
L5.VRPx					0.38 (1.56)	0.16 (0.72)	0.25 (0.82)	0.25 (0.83)
L5.VRPx _{PD}						1.06*** (3.44)		1.16** (2.55)
L5.VRPx _{REC}							0.22 (0.47)	-0.19 (-0.37)
R^2	0.000	0.030	0.014	0.075	0.008	0.022	0.009	0.023

Newey-West t statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table includes coefficients from predictability regressions with the market excess return multiplied by 100 as the dependent variable. The independent variables (with a lag of four or five months) are the variance risk premium (VRP) computed from SVIX as in Martin (2017), and the variance risk premium computed using VIX instead of SVIX (denoted VRPx), VRP_{PD}, which is the product of VRP and an indicator function that equals 1 for periods in the lowest decile of PD values (VRPx_{PD} is defined analogously to VRP_{PD}), VRP_{REC}, which is the product of VRP and an indicator function that equals 1 for NBER recession periods (VRPx_{REC} is defined analogously to VRP_{REC}). To aid in interpreting the coefficients, VRP and VRPx are each divided by their sample (monthly) standard deviations, 0.00251 and 0.00249, respectively.

Table 3. Return Predictability of Crash Probability and Implied Correlation in Low PD Periods

	(1) MKTRF%	(2) MKTRF%	(3) MKTRF%	(4) MKTRF%	(5) MKTRF%	(6) MKTRF%	(7) MKTRF%	(8) MKTRF%
L4.CP	0.66** (2.13)	-0.09 (-0.15)	0.98** (2.06)	0.77* (1.73)				
L4.CP _{PD}		1.40** (2.43)		3.20*** (4.95)				
L4.CP _{REC}			-0.45 (-0.71)	-2.60*** (-3.34)				
L5.CP					0.90*** (2.93)	0.53 (1.33)	0.81* (1.69)	0.74 (1.55)
L5.CP _{PD}						0.70** (2.09)		1.14 (1.52)
L5.CP _{REC}							0.13 (0.27)	-0.64 (-0.82)
R^2	0.019	0.046	0.022	0.099	0.034	0.041	0.034	0.044
	(1) MKTRF%	(2) MKTRF%	(3) MKTRF%	(4) MKTRF%	(5) MKTRF%	(6) MKTRF%	(7) MKTRF%	(8) MKTRF%
L4.IC	0.69*** (2.96)	0.53** (2.06)	0.77*** (3.30)	0.62*** (2.69)				
L4.IC _{PD}		1.03*** (3.74)		2.08*** (3.36)				
L4.IC _{REC}			-0.42 (-0.90)	-1.14*** (-2.27)				
L5.IC					0.57** (2.23)	0.44** (1.82)	0.64** (2.40)	0.51** (2.05)
L5.IC _{PD}						0.90*** (4.62)		1.73*** (3.87)
L5.IC _{REC}							-0.31 (-0.64)	-0.91** (-2.12)
R^2	0.024	0.047	0.034	0.097	0.017	0.034	0.022	0.066

Newey-West t statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table includes coefficients from predictability regressions with the market excess return multiplied by 100 as the dependent variable. The independent variables (with a lag of four or five months) are the option-implied crash probability (CP) from Martin (2017), and the option-implied correlation (IC) from Buss et al. (2019), CP_{PD}, which is the product of CP and an indicator function that equals 1 for periods in the lowest decile of PD values (IC_{PD} is defined analogously to CP_{PD}), and CP_{REC}, which is the product of CP and an indicator function that equals 1 for NBER recession periods (IC_{REC} is defined analogously to CP_{REC}). To aid in interpreting the coefficients, CP and IC are each divided by their sample (monthly) standard deviations, 0.011 and 0.132, respectively.

Table 4. Return Predictability of the Pricing Kernel U-shape

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%
L4. $\tilde{M}$	0.09 (0.42)	-0.03 (-0.11)	0.14 (0.62)	0.14 (0.63)				
L4. $\tilde{M}_{PD}$		2.95*** (5.14)		3.96*** (3.94)				
L4. $\tilde{M}_{REC}$			-0.26 (-0.32)	-1.17 (-1.55)				
L5. $\tilde{M}$					-0.02 (-0.12)	-0.09 (-0.51)	0.03 (0.19)	0.04 (0.20)
L5. $\tilde{M}_{PD}$						1.65*** (6.54)		2.38*** (3.20)
L5. $\tilde{M}_{REC}$							-0.30 (-0.51)	-0.85 (-1.19)
R^2	0.000	0.029	0.001	0.042	0.000	0.009	0.001	0.016

Newey-West t statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table includes coefficients from predictability regressions with the market excess return multiplied by 100 as the dependent variable. The independent variables (with a lag of four or five months) are the increasing portion of the right tail of the pricing kernel ($\tilde{M}$) from equation (5) in Sichert (2020) with the physical distribution from Bekaert and Hoerova (2014), $\tilde{M}_{PD}$, which is the product of $\tilde{M}$ and an indicator function that equals 1 for periods in the lowest decile of PD values, and $\tilde{M}_{REC}$, which is the product of $\tilde{M}$ and an indicator function that equals 1 for NBER recession periods. To aid in interpreting the coefficients, $\tilde{M}$ is divided by its standard deviations, 4.67.

5. The Pricing Kernel Puzzle

The pricing kernel puzzle (Cuesdeanu and Jackwerth, 2018) is the empirical finding that the pricing kernel or stochastic discount factor (SDF) is not monotonically decreasing in the market return, but rather increases at the right tail of the distribution, generating a U-shape (Bakshi et al., 2010; Sichert, 2020). The NEO-EU representative agent generates a U-shaped pricing kernel with a more pronounced U-shape in times of high optimism and in times of high uncertainty. More specifically, the extra weight on the optimistic state, $\alpha_t \gamma_t$, generates a U-shaped pricing kernel. This follows, because a higher perceived probability of the optimistic state increases the price of a state-contingent claim that pays in the optimistic scenario, resulting in an upward tick at the right end of the pricing kernel. Consequently, we have the following prediction:

PROPOSITION 1: *In the Lucas-tree environment with a NEO-EU representative agent with a CRRA utility function, the pricing kernel has a U-shape in returns. Moreover, the increasing part of the pricing kernel is increasing in $\alpha\gamma$ (the U-shape is more pronounced when $\alpha\gamma$ is high). Further, for a persistent (transitory) process for α (γ), variation in α drives long-run variation in the pricing kernel, whereas variation in γ drives short-run variation in the pricing kernel.*

Proposition 1 is in contrast to standard asset pricing models including habit formation (Campbell and Cochrane, 1999), long-run risk (Bansal and Yaron, 2004), and rare disasters (Wachter, 2013) that produce a monotonically decreasing pricing kernel when projecting the pricing kernel onto the market return, and hence, these models do not generate the U-shape or its dynamics.

In the Lucas-tree environment, the consumption growth rate approximately equals the wealth growth rate that in turn approximately equals the market return. As a result, the pricing kernel with the NEO-EU representative agent in terms of the market return has a U-shape.

For purely expositional purposes, we observe that the U shape of the pricing kernel is expressed most simply in a setting with a finite number of states. In that case, replacing the consumption growth with the market return in the NEO-EU agent stochastic discount factor, M_{t+1}^{NEU} , and continuing to assume that the representative agent truncates the support at $[\underline{R}, \bar{R}]$ yields

$$M_{t+1}^{NEU} = \begin{cases} (1 - \gamma_t)\beta \underline{R}_{t+1}^{-\theta} + \gamma_t(1 - \alpha_t)\beta \underline{R}_{t+1}^{-\theta} & \text{if } R_{t+1} = \underline{R}_{t+1} \\ (1 - \gamma_t)\beta \underline{R}_{t+1}^{-\theta} & \text{if } R_{t+1} \in (\underline{R}_{t+1}, \bar{R}_{t+1}) \\ (1 - \gamma_t)\beta \bar{R}_{t+1}^{-\theta} + \gamma_t\alpha_t\beta \bar{R}_{t+1}^{-\theta} & \text{if } R_{t+1} = \bar{R}_{t+1}. \end{cases} \quad (10)$$

The weights above clearly show that the pricing kernel is U-shaped in R . Moreover, a higher $\alpha_t \gamma_t$ indicates a larger increase (jump) on the right side of the pricing kernel.

Fortunately, the literature has recently developed empirical measures for the increasing part of the pricing kernel, which we denote with $\tilde{M}$.²⁰ Let us start with the null of Proposition 1,

²⁰Sichert measures the right tail of the stochastic discount factor as the ratio of the cumulative density of the risk-neutral distribution that is not covered by his (cleaned) option data to the cumulative density of the physical distribution. In particular, he employs the formula $\hat{M}_{t,t+\tau}^{righttail} = (1 - F^*(R_{t,t+\tau}^{max})) / (1 - \hat{F}(R_{t,t+\tau}^{max}))$, where R^{max} is

and assuming that BW and VRP are valid (but noisy) proxies for α and γ , respectively. Then, a current high $\tilde{M}$ could be either due to a high α (BW), or high γ (VRP), or something else (q , etc). Importantly, the future behavior of $\tilde{M}$ depends on the source of its current shock: if $\tilde{M}$ is high because γ is high, then this effect should disappear after several months (since γ shocks are transitory). However, if $\tilde{M}$ is high due to a high α , then its effect should remain even after a couple of years due to the high persistence of α . Thus, we expect to see a significant positive correlation between BW_t and $\tilde{M}_{t+h}$, and more importantly, we expect the correlation to increase with h . This is because other contaminating contemporaneous (but transitory) factors in $\tilde{M}_{t+h}$ keep dying out except for the α_t signal present in BW_t . Table 5 reports the correlation coefficients between the

	$\tilde{M}_{t+3}$	$\tilde{M}_{t+6}$	$\tilde{M}_{t+9}$	$\tilde{M}_{t+12}$	$\tilde{M}_{t+15}$	$\tilde{M}_{t+18}$	$\tilde{M}_{t+21}$
BW_t	0.14 ^{**}	0.17 ^{***}	0.26 ^{***}	0.30 ^{***}	0.39 ^{***}	0.46 ^{***}	0.48 ^{***}
ICS_t	-0.01	-0.01	0.04	0.09	0.11 [*]	0.12 ^{**}	0.20 ^{***}

Table 5. Sentiment indexes predicting future right tail of the SDF.

current BW index and the future $\tilde{M}$'s at quarterly horizons, between 3 months and 21 months. Note that the correlations are all positive and significant, and their significance increases with the horizon. The magnitudes are also large: BW predicts over 20% of the long-run variation in the right tail of the SDF (with an adjusted R^2 of 22% at the two-year horizon). The second row uses the Michigan index of consumer sentiment (ICS) for comparison, although it is survey-based and arguably a poorer proxy for α_t compared to BW_t . However, the same pattern is clearly visible, only at longer horizons and with a lower magnitude. Indeed, this pattern exists for other measures of sentiment as well, which we do not report for brevity. We want to emphasize that this finding is new in the literature, and shows a strong link between the sentiment indexes and the right tail of the stochastic discount factor, which is associated with the implied perceived probability of highest returns. Although not shown in Table 5, the significance levels gradually drop after the second year but remain significant for some time.

To further test the predictions of Proposition 1, we repeat the previous exercise but with VRP instead of BW. In this case, we also expect to see a positive and significant correlation between the current VRP (proxy of γ) and $\tilde{M}_{t+h}$. However, after some initial periods, we expect that the significance of the correlations drop. This is because shocks to γ (and VRP) are transitory, and although other contaminating shocks disappear after a few periods, the signal in γ_t itself should fade relatively fast compared to a signal in α_t . Table 6 reports the correlations between VRP_t and $\tilde{M}_{t+h}$ for quarterly horizons between 3 months and 21 months. As expected, only the first three quarters have positive and significant correlations. The significance sharply drops to zero after

the return corresponding to the highest available strike price that passes his data filters, and $\hat{F}$ is the cumulative density function corresponding to the physical distribution. As our analysis uses only the right tail of the stochastic discount factor, we use the simpler notation: $\tilde{M} \equiv \hat{M}_{t,t+\tau}^{righttail}$. Empirically, Sichert constructs two versions of this measure, using different measures of the physical distribution. Although we employ one of these measures in our analysis, our empirical results in this section and for pricing kernel return predictability from Section 4 hold for both versions of Sichert's measure.

	$\tilde{M}_{t+3}$	$\tilde{M}_{t+6}$	$\tilde{M}_{t+9}$	$\tilde{M}_{t+12}$	$\tilde{M}_{t+15}$	$\tilde{M}_{t+18}$	$\tilde{M}_{t+21}$
VRP _t	0.18**	0.23***	0.20***	0.12	0.02	0.10	-0.07
EPU _t	0.25***	0.15**	0.12*	0.08	0.07	0.10	-0.08

Table 6. Uncertainty indexes predicting future right tail of the SDF.

three quarters. The second row of Table 6 reports the correlations between the current measure of Economic Policy Uncertainty (EPU_t) and $\tilde{M}_{t+h}$ as a robustness check. Note that the pattern is very similar, but this time the highest correlation happens after one quarter and the significance drops to zero after three quarters.

In summary, Tables 5 and 6 not only provide new empirical evidence that the increasing part of the pricing kernel positively depends on sentiment and uncertainty proxies, and this positive relation is consistent with a slow moving signal in sentiment and a fast moving signal in uncertainty, but also it provides a direct link that sentiment indices are related to right tail events, and they are not just about the mean of the return distribution.

6. The Mean-Variance Puzzle

The risk-return tradeoff (whether market volatility positively predicts excess returns) has been investigated and debated in many empirical studies.²¹ One might expect that such a fundamental relation in finance would be clearly and robustly present in the data. Rational risk-averse investors should require additional compensation in equilibrium for holding stocks during riskier periods (*i.e.*, periods with higher market volatility). Such a relationship is predicted by standard asset pricing models as shown by Merton (1980). Yet, as noted by Yu and Yuan (2011), “numerous studies over the past three decades find rather mixed empirical evidence of such a relation” (p.367). Yu and Yuan (2011) reconcile the evidence by finding that the slope of the mean-variance frontier is flat (steep) at times of high (low) sentiment.

The empirical finding of Yu and Yuan (2011) can be analytically derived as an implication from the NEO-EU CAPM, as shown in the approximation in Proposition 2. In particular, when market optimism is not too high ($\alpha_t < \frac{\xi}{\xi+\bar{\xi}}$) the expected excess return is increasing in the conditional market volatility (q_t), but this slope flattens as market optimism, α_t , rises.

Using the exponential approximation $e^x \approx 1 + x$, and the notation $r_{f,t} := \ln(1 + r_t^f)$, the equity premium reduces, up to the first order approximation, to a formula that depends only on state variables q_t, γ_t , and α_t and parameters ξ and $\bar{\xi}$. These variables generate a time-varying risk-premium (due to q_t), uncertainty premium (due to γ_t), and sentiment premium (due to α_t):

²¹French et al. (1987); Baillie and DeGennaro (1990); Campbell and Hentschel (1992); Ghysels et al. (2005); Lundblad (2007); Guo and Whitelaw (2006); Brandt and Wang (2007); Pástor et al. (2008) find a positive risk-return tradeoff, while Campbell (1987); Nelson (1991); Whitelaw (1994); Brandt and Kang (2004); Lettau and Ludvigson (2010) find a negative risk-return relation.

PROPOSITION 2: *The equity premium under the NEO-EU CAPM is approximately given by the following decomposition into risk and speculation components.*

$$E_t r_{t+1} - r_{f,t} \approx \underline{\xi} \gamma_t q_t - (\underline{\xi} + \bar{\xi}) \alpha_t \gamma_t q_t.$$

The risk component, $\underline{\xi} \gamma_t q_t$, is proportional to the product of market uncertainty, γ_t , and market volatility, q_t . The speculation component, $-(\underline{\xi} + \bar{\xi}) \alpha_t \gamma_t q_t$, depends further on market optimism, α_t .

Proof. See Appendix B. □

Proposition 2 also provides a functional form for the expected equity premium that can be tested with the sentiment and uncertainty proxies. We construct two factors to represent the time-varying components in Proposition 2. We construct the first factor as the product of VRP_t and q_t (which is our proxy for the $\gamma_t q_t$ factor in the proposition). Next, we note that the HJTZ sentiment index of Huang et al. (2015) is constructed from the same proxies as BW, but it was specifically constructed to better predict aggregate market returns.²² We construct the second factor as the product of $HJTZ_t$, VRP_t , and q_t (which is our proxy for the $\alpha_t \gamma_t q_t$ factor in the proposition).

We run predictability regressions of our proxies for the two factors in Proposition 2, to test if they predict the realized equity premium, MKTRF. The results are shown in Table 7 for a lag of 4 months, which was also used in our predictability regressions in Section 4. As can be seen from the table, in univariate predictability regressions, the product of VRP and q does not predict the realized equity premium. This is consistent with the difficulty of detecting a significant risk-return trade-off in the empirical literature. Moreover, in univariate predictability regressions with VRP, q , or HJTZ we find that only HJTZ predicts the realized equity premium at a four-month lag unconditionally and does so with a Newey-West t-statistic of -2.45, lower than that of the factors $H*VRP*q$ and $H*VRP_x*q$ in the second and fourth regressions in Table 7. When both factors from Proposition 2 are included as suggested by the proposition, both factors significantly predict the equity premium with the expected signs.

Accounting for an optimism factor thus helps to identify the relationship between the equity premium and market volatility. While this empirical finding is consistent with Yu and Yuan (2011), our approach goes further in three respects: (i) Whereas Yu and Yuan link the equity premium to market volatility and sentiment, our approach also links Knightian uncertainty to these variables; (ii) We derive the result as a theoretical implication from a version of a standard model of choice under uncertainty in decision theory; (iii) We use the theoretical result to obtain a precise functional form relating the state variables and find it predicts the realized equity premium.

More recently, Brenner and Izhakian (2018) find that accounting for ambiguity and ambiguity attitude helps explain the risk-return tradeoff. There are two important differences compared to our approach. First, Brenner and Izhakian (2018) do not have a role for time-varying uncertainty

²²The HJTZ index is constructed from the sentiment proxies using partial least squares, whereas BW is constructed using principal components analysis. Huang et al. (2015) emphasize that their index is “aligned with the purpose of predicting the aggregate stock market” and that it is a “powerful predictor of stock returns.”

attitude which plays a key role in our setting in explaining time variation in the risk-return tradeoff. Second, we derive a closed form relationship between expected excess returns, market volatility, market uncertainty, and market optimism in the framework of a standard axiomatic model for choice under uncertainty that makes the relationship between these variables transparent.

Table 7. Optimism, Uncertainty, and the Mean-Variance Relation

	MKTRF%	MKTRF%		MKTRF%	MKTRF%
L4.VRP* q	0.33 (0.79)	0.81** (2.45)	L4.VRP x^*q	0.22 (0.71)	0.60** (2.40)
L4.H*VRP* q		-0.91*** (-3.00)	L4.H*VRP x^*q		-0.74*** (-3.35)
R ²	0.005	0.031		0.003	0.025

Newey-West t statistics in parentheses. ** $p < 0.05$, *** $p < 0.01$

Notes: The table displays predictability regressions of the realized equity premium, MKTRF, from CRSP, multiplied by 100. The predictors (with a four-month lag) are (i) the product of the variance risk premium (constructed from SVIX in the two leftmost regressions, constructed from VIX in the two rightmost regressions) and market volatility (VRP* q) and (ii) the product of VRP, q , and the HJTZ sentiment index from Huang et al. (2015) (H*VRP* q). To aid with interpreting the coefficients, each predictor is divided by its standard deviation. The two leftmost regressions span our full sample of SVIX data (January, 1996 through January, 2012). The two rightmost regressions span our full sample period (January, 1990 through December, 2018).

7. Discussion

Here, we briefly discuss two broader themes of the paper: (i) identifying market risk, market uncertainty, and market optimism as state variables and (ii) distinguishing the ‘Knightian uncertainty’ and ‘rare events’ interpretations of the model.

7.1. Market Risk, Uncertainty, and Optimism as State Variables

We consider the case for interpreting market volatility (q), market uncertainty (γ), and market optimism (α) as state variables for the aggregate market that are subject to shocks. To do so, we summarize potentially appealing or important properties for state variables to have in a representative agent asset pricing model and we discuss how α , γ , and q relate to these properties.

1. **The state variables predict time varying expected excess returns:** We showed in Section 6 that proxies for the state variables jointly predict the realized equity premium.
2. **The state variables are sources of systematic risk:** In the CAPM, market volatility (q) is the only risk priced in equilibrium. In the present approach, three sources of systematic

risk (market volatility, market uncertainty, and market optimism) are priced.²³

3. **The state variables predict changes in the investment opportunity set:** Market volatility, uncertainty, and optimism each plausibly affect the investment opportunity set, and so can be state variables in the sense of Merton's ICAPM (Merton, 1973) framework.
4. **The state variables summarize the state of the market:** Regime switching models study behavior across different market states. Prominent studies have employed regime switching models with a high and low volatility regime (Baele, 2005), or a high and low uncertainty regime (Bianchi et al., 2018), or a high and low sentiment regime (Yu and Yuan, 2011). This suggests that market volatility, market uncertainty, and market optimism are each natural and standard approaches to classifying and summarizing the state of the market.
5. **At least two state variables drive time-varying expected excess returns:** Schlag et al. (2021) find that “A model in which only one state variable causes expected excess returns to be time-varying ... is by construction not able to match the cross-sectional patterns in average returns in the data. Specifying a framework where at least two state variables drive the time variation in expected excess returns is thus a necessary, but still not sufficient, condition to explain the empirical stylized facts.”
6. **One state variable is positively related to the market price dividend ratio:** Schlag et al. (2021) conclude, “the two state variables have to exhibit very specific characteristics to allow the model to match the data. Most importantly, one of the state variables has to be positively related to the price-dividend ratio of the aggregate market. This feature is rare in asset pricing models building on endowment economies, because state variables driving the time variation in risk premia, like risk aversion, uncertainty, or disaster risk, typically have the property that an increase in any of them leads to lower valuation ratios.” In our setting, market optimism, α , has the key property that it is positively related to PD.

7.2. Knightian Uncertainty versus Rare Events

The representative agent's problem in (2) is a special case of two major paradigms for modeling choice under uncertainty (based on multiple priors or Choquet capacities). Moreover, α and γ have established interpretations in the decision theory literature as reflecting uncertainty attitude and perceived uncertainty, and these are viewed as personal characteristics of a decision maker that are elicited in experimental studies (*e.g.*, Baillon et al., 2018; Dimmock et al., 2015).

The foundations for the representative agent in (2) extend beyond the uncertainty literature. Although the uncertainty preference interpretation is natural and well-established for the representative agent in (2), note that this agent also has an observationally equivalent interpretation

²³These state variables result in a risk premium (associated with q), an uncertainty premium (associated with γ , in which investors demand higher returns in equilibrium for assets more sensitive to uncertainty, if optimism is not too high), and an optimism discount (associated with α , in which investors accept lower returns in equilibrium for assets more sensitive to market optimism).

as an expected utility maximizing agent with stable preferences in an economy with time-varying rare event probabilities. In particular, the model could be interpreted as capturing three regimes - ‘normal’ times which occur with (objective) probability $(1 - \gamma_t)$, a rare disaster or bear market state that occurs with probability $\gamma_t(1 - \alpha_t)$, and a rare boom or bull market state that occurs with probability $\gamma_t\alpha_t$. The representative agent in (2) thereby unifies models based on Knightian uncertainty and models based on rare events.²⁴

There are three important differences, however, between the Knightian uncertainty and rare events interpretations. First, the rare events interpretation parameterizes the model to have a disaster probability and a boom probability (the products $\gamma(1 - \alpha)$ and $\gamma\alpha$, from the perspective of the uncertainty interpretation). *Regardless of what α and γ represent*, the information pertaining to the separate dynamics of α and γ (the high persistence of α and the low persistence of γ) is missed when they are combined into boom and disaster probabilities. As we have shown in Sections 4 and 5, the separate dynamics of α and γ are important in unifying the return predictability of PD, VRP, CP, IC, and $\tilde{M}$, and in explaining the short and long run variation in the shape of the stochastic discount factor. In contrast, parameterizing the model state variables as boom and disaster probabilities effectively combines α and γ and hence, conceals their separate dynamics.

Second, α and γ , representing market optimism and market uncertainty, *are themselves conceptually important*, as they offer the potential to connect equilibrium asset pricing theory to the large literature on empirical proxies for market sentiment and uncertainty. As we have shown in this paper, α and γ can be used to generate new testable predictions regarding standard empirical measures of market optimism (such as the Baker-Wurgler Baker and Wurgler (2006) index of market sentiment) and market uncertainty (such as the variance risk premium). We find that these predictions from theory are consistently observed in the data. In this respect, α and γ could help to organize the list of empirical findings documented for measures of market optimism and market uncertainty which otherwise lack a general analytical framework to predict their behavior in applications. These predictions are missed when the state variables are boom and disaster probabilities that reflect rational expectations, as in the rare events framework, which removes the concepts of sentiment and Knightian uncertainty from the model and thereby restricts the scope of the theory.

Third, under the rare events interpretation, agents have rational expectations, the market is always efficient, and all assets are priced correctly. In contrast, the Knightian uncertainty interpretation opens an avenue to study systematic mispricing within equilibrium asset pricing theory. For instance, the equity premium formula in (5) can be viewed as consisting of the risk premium in an efficient market (if one treats the CCAPM as a reasonable benchmark for efficient pricing) plus a mispricing component that varies systematically across assets and over time. The model can then be used to study time variation in market efficiency and mispricing, consistent with recent empirical work which argues that market efficiency is not static, but varies systematically across assets (Stambaugh and Yuan, 2017) and over time (Rösch et al., 2017).

²⁴Rare positive events are referred to as bonanzas in Barro and Liao (2021), and as booms in Tsai and Wachter (2016). The latter paper studies a continuous time model with rare booms and disasters in order to explain the value premium in the cross-section of returns.

All three differences between the rare events interpretation (with time-varying boom and disaster probabilities) and the Knightian uncertainty interpretation (with time-varying optimism and uncertainty) also apply to distinguish the Knightian uncertainty interpretation from models with good and bad uncertainties (Bekaert and Engstrom, 2017; Segal et al., 2015).

8. Conclusion

Four basic types of asset pricing puzzles for the aggregate stock market are (i) puzzles arising from the difficulty in matching basic stock market moments (such as the equity premium puzzle), (ii) predictability puzzles arising from the empirical evidence on stock return predictability (such as the predictability of the price-dividend ratio), (iii) the pricing kernel puzzle arising from the non-monotonicity of the pricing kernel and its tendency to follow a U shape, and (iv) the difficulty in identifying a positive risk-return tradeoff for the aggregate market. Much of the earlier literature on these types of puzzles has focused on matching the equity premium, explaining the predictability of PD, explaining the U-shape of the pricing kernel, or finding alternative ways to measure the risk-return tradeoff. The present paper further advances this literature by simultaneously matching four basic market moments (the risk-free rate, the equity premium, the variance risk premium, and the risk-neutral entropy), by providing a unified explanation for the return predictability of five aggregate market variables (PD, VRP, CP, IC, and $\tilde{M}$) that we test empirically, by providing an explanation for the time variation in the U-shape of the pricing kernel that we test empirically, and by providing a theoretical result relating the equity premium to market volatility, market uncertainty, and market optimism, that we test empirically. In each empirical test, we observe strong support for the model predictions. For each puzzle, we find that the disciplined deviation from rational expectations formalized by NEO-EU preferences adds to the current state of knowledge.

For each type of puzzle we identified an important role for optimistic attitudes toward Knightian uncertainty. By overweighting the right tail of the risk-neutral distribution, market optimism helps match the risk-neutral variance of the excess market return and the market risk-neutral entropy. The high persistence of market optimism interacts with the low persistence of market uncertainty to generate a new mechanism for return predictability. The same properties of market optimism and uncertainty help explain the long-run variation in the shape of the pricing kernel. Time variation in market optimism further helps to identify the risk-return tradeoff for the aggregate market.

The calibration that matches the VIX-SVIX difference, the new return predictability mechanism, the new source of long-run variation in the pricing kernel U-shape, and the identification of the risk-return tradeoff would have been missed if one did not allow for a positive level of market optimism. In evaluating the evidence of whether there is a benefit to incorporating optimistic uncertainty attitudes into asset pricing models, we thus conclude there is a strong case for optimism.

Appendix A Stock Market Moments

This section contains the details of the derivation of the risk-free rate, equity premium, risk-neutral entropy, and risk-neutral variance of the excess market return as functions of the model state variables for Knightian uncertainty (γ_t), and uncertainty attitude (α_t), together with the conditional expectations of market log return (x_t), and its volatility (q_t).

Recall from Section 3.3.1 that if $E_t h_{t+1}$ is linear in h_t , equations (6) and (7) imply that $E_t r_{t+1} = \theta_0 + \theta_1 p d_t$, where θ_0 and θ_1 are constants. Moreover, our conditional Normality assumption implies

$$\begin{bmatrix} r_{t+1} \\ \Delta c_{t+1} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} x_t \\ \mu \end{bmatrix}, \begin{bmatrix} q_t^2 & q_t \sigma \eta \\ q_t \sigma \eta & \sigma^2 \end{bmatrix} \right),$$

where $r_{t+1} := \log(R_{t+1})$, $\Delta c_{t+1} := \log(\frac{C_{t+1}}{C_t})$, and $x_t := \theta_0 + \theta_1 p d_t$. Thus, our truncation assumption implies that

$$\begin{aligned} \overline{\Delta c_{t+1}} &= \mu + \bar{\xi} \sigma, & \overline{r_{t+1}} &= x_t + \bar{\xi} q_t \\ \underline{\Delta c_{t+1}} &= \mu - \underline{\xi} \sigma, & \underline{r_{t+1}} &= x_t - \underline{\xi} q_t, \end{aligned}$$

where $\underline{\xi}$ and $\bar{\xi}$ are fixed parameters.

Risk-free return. We start with Equation (4), which is the Euler equation for the gross risk-free return $R_{f,t}$ (reproduced below for convenience).

$$(1 - \gamma_t) E_t M_{t+1} R_{f,t} + \gamma_t (\alpha_t \overline{M}_{t+1} R_{f,t} + (1 - \alpha_t) \underline{M}_{t+1} R_{f,t}) = 1,$$

where $M_{t+1} := \beta \frac{u'(C_{t+1})}{u'(C_t)}$, and $\overline{M}_{t+1}$ and $\underline{M}_{t+1}$ are associated with the next period's optimistic and pessimistic consumption growth rates. Given the CRRA utility function, $M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\theta} = e^{-\rho - \theta \Delta c_{t+1}}$, where $\rho := -\ln \beta$. Then, the normality of Δc_{t+1} implies $E_t M_{t+1} = e^{-\rho - \theta \mu + \frac{1}{2} \theta^2 \sigma^2}$, and our truncation assumption implies $\overline{M}_{t+1} = e^{-\rho - \theta \mu - \theta \bar{\xi} \sigma}$, and $\underline{M}_{t+1} = e^{-\rho - \theta \mu + \theta \underline{\xi} \sigma}$. Thus, we have

$$R_{f,t} = \frac{1}{(1 - \gamma_t) e^{-\rho - \theta \mu + \frac{1}{2} \theta^2 \sigma^2} + \gamma_t (\alpha_t e^{-\rho - \theta(\mu + \bar{\xi} \sigma)} + (1 - \alpha_t) e^{-\rho - \theta(\mu - \underline{\xi} \sigma)})}. \quad (11)$$

Equity premium. We start with the Euler equation (3) that is reproduced below for convenience.

$$(1 - \gamma_t) E_t M_{t+1} R_{t+1} + \gamma_t (\alpha_t \overline{M}_{t+1} \overline{R}_{t+1} + (1 - \alpha_t) \underline{M}_{t+1} \underline{R}_{t+1}) = 1.$$

Subtracting the Euler equation with the gross risk-free return from the above, then substituting $E_t M_{t+1} R_{t+1}$ with $E_t M_{t+1} E_t R_{t+1} + \text{Cov}_t(M_{t+1}, R_{t+1})$, and then rearranging terms and dividing by

$(1 - \gamma_t)E_t M_{t+1}$ to have the equity premium $EP_t := E_t R_{t+1} - R_{f,t}$ on the left hand side, yields

$$EP_t = -\frac{\text{Cov}_t(M_{t+1}, R_{t+1})}{E_t M_{t+1}} + \frac{\gamma_t}{1 - \gamma_t} \left(\alpha_t \frac{\bar{M}_{t+1}}{E_t M_{t+1}} (R_{f,t} - \bar{R}_{t+1}) + (1 - \alpha_t) \frac{\underline{M}_{t+1}}{E_t M_{t+1}} (R_{f,t} - \underline{R}_{t+1}) \right).$$

The rest of the calculation is straightforward but we provide it below for the readers' convenience. Starting with the covariance term, we have

$$\begin{aligned} \text{Cov}_t(M_{t+1}, R_{t+1}) &= E_t M_{t+1} R_{t+1} - E_t M_{t+1} E_t R_{t+1} = E_t e^{-\rho - \theta \Delta c_{t+1} + r_{t+1}} - E_t e^{-\rho - \theta \Delta c_{t+1}} E_t e^{r_{t+1}} \\ &= e^{-\rho - \theta \mu + \frac{1}{2} \theta^2 \sigma^2 + x_t + \frac{1}{2} q_t^2} (e^{-\theta \text{Cov}_t(\Delta c_{t+1}, r_{t+1})} - 1) \\ &= e^{-\rho - \theta \mu + \frac{1}{2} \theta^2 \sigma^2 + x_t + \frac{1}{2} q_t^2} (e^{-\theta \eta \sigma q_t} - 1). \end{aligned}$$

η is the time invariant correlation between the log market return and log consumption growth.²⁵ Substituting $E_t M_{t+1}$, $\bar{M}_{t+1}$ and $\underline{M}_{t+1}$ with their model equivalents from the previous part, we get

$$\begin{aligned} EP_t &= e^{x_t + 0.5 q_t^2} (e^{-\eta \theta \sigma q_t} - 1) + \\ &\quad \frac{\gamma_t \alpha_t}{1 - \gamma_t} e^{-\theta \bar{\xi} \sigma - 0.5 \theta^2 \sigma^2} (R_{f,t} - e^{x_t + \bar{\xi} q_t}) + \frac{\gamma_t (1 - \alpha_t)}{1 - \gamma_t} e^{\theta \underline{\xi} \sigma - 0.5 \theta^2 \sigma^2} (R_{f,t} - e^{x_t - \underline{\xi} q_t}). \end{aligned} \quad (12)$$

VIX. From Martin (2017), we have $VIX_t^2 = 2(\log \tilde{E}_t^Q R_{t+1} - \tilde{E}_t^Q \log R_{t+1})$ from t to $t + 1$, where Q denotes the risk-neutral measure. Recall that $\tilde{E}_t^Q R_{t+1} := \frac{\tilde{E}_t M_{t+1} R_{t+1}}{\tilde{E}_t M_{t+1}}$. Substituting in the numerator and the denominator, and then removing the common factor $e^{-\rho - \theta \mu}$ yields

$$\tilde{E}_t^Q R_{t+1} = \frac{(1 - \gamma_t) e^{x_t + 0.5(\theta^2 \sigma^2 + q_t^2 - 2\eta \theta \sigma q_t)} + \gamma_t \left(\alpha_t e^{-\theta \bar{\xi} \sigma + x_t + \bar{\xi} q_t} + (1 - \alpha_t) e^{\theta \underline{\xi} \sigma + x_t - \underline{\xi} q_t} \right)}{(1 - \gamma_t) e^{0.5 \theta^2 \sigma^2} + \gamma_t \left(\alpha_t e^{-\theta \bar{\xi} \sigma} + (1 - \alpha_t) e^{\theta \underline{\xi} \sigma} \right)}.$$

To find $\tilde{E}_t^Q \log R_{t+1} = \frac{\tilde{E}_t M_{t+1} \log R_{t+1}}{\tilde{E}_t M_{t+1}} = \frac{\tilde{E}_t e^{-\rho - \theta \Delta c_{t+1}} r_{t+1}}{\tilde{E}_t e^{-\rho - \theta \Delta c_{t+1}}}$, we use $E e^X Y = e^{\mu_x + 0.5 \sigma_x^2} (\mu_y + \eta_{x,y} \sigma_x \sigma_y)$, where E is expected value, and X and Y have a bivariate Normal distribution with means μ_x and μ_y , variances σ_x^2 and σ_y^2 , and a correlation coefficient of $\eta_{x,y}$. Thus,

$$\begin{aligned} VIX_t^2 &= 2 \left\{ \log \left[\frac{(1 - \gamma_t) e^{x_t + 0.5(\theta^2 \sigma^2 + q_t^2 - 2\eta \theta \sigma q_t)} + \gamma_t \left(\alpha_t e^{-\theta \bar{\xi} \sigma + x_t + \bar{\xi} q_t} + (1 - \alpha_t) e^{\theta \underline{\xi} \sigma + x_t - \underline{\xi} q_t} \right)}{(1 - \gamma_t) e^{0.5 \theta^2 \sigma^2} + \gamma_t \left(\alpha_t e^{-\theta \bar{\xi} \sigma} + (1 - \alpha_t) e^{\theta \underline{\xi} \sigma} \right)} \right] - \right. \\ &\quad \left. \frac{(1 - \gamma_t) e^{0.5 \theta^2 \sigma^2} (x_t - \eta \theta \sigma q_t) + \gamma_t \left(\alpha_t e^{-\theta \bar{\xi} \sigma} (x_t + \bar{\xi} q_t) + (1 - \alpha_t) e^{\theta \underline{\xi} \sigma} (x_t - \underline{\xi} q_t) \right)}{(1 - \gamma_t) e^{0.5 \theta^2 \sigma^2} + \gamma_t \left(\alpha_t e^{-\theta \bar{\xi} \sigma} + (1 - \alpha_t) e^{\theta \underline{\xi} \sigma} \right)} \right\}. \end{aligned} \quad (13)$$

²⁵ A multi-variable GARCH model suggests that this correlation is indeed time-invariant.

SVIX. Martin (2017) defines $\text{SVIX}_t^2 = \text{Var}_t^Q\left(\frac{R_{t+1}}{R_{f,t}}\right)$ from t to $t+1$, where Q denotes the risk-neutral measure. In the context of our model, we have $\text{Var}_t^Q = \tilde{E}_t^Q R_{t+1}^2 - (\tilde{E}_t^Q R_{t+1})^2$. Thus,

$$\text{SVIX}_t^2 = \frac{1}{R_{f,t}^2} \left\{ \frac{(1-\gamma_t)e^{2x_t+0.5(\theta^2\sigma^2+4q_t^2-4\eta\theta q_t\sigma)} + \gamma_t \left(\alpha_t e^{-\theta\bar{\xi}\sigma+2(x_t+\bar{\xi}q_t)} + (1-\alpha_t)e^{\theta\underline{\xi}\sigma+2(x_t-\underline{\xi}q_t)} \right)}{(1-\gamma_t)e^{0.5\theta^2\sigma^2} + \gamma_t \left(\alpha_t e^{-\theta\bar{\xi}\sigma} + (1-\alpha_t)e^{\theta\underline{\xi}\sigma} \right)} - \left[\frac{(1-\gamma_t)e^{x_t+0.5(\theta^2\sigma^2+q_t^2-2\eta\theta\sigma q_t)} + \gamma_t \left(\alpha_t e^{-\theta\bar{\xi}\sigma+x_t+\bar{\xi}q_t} + (1-\alpha_t)e^{\theta\underline{\xi}\sigma+x_t-\underline{\xi}q_t} \right)}{(1-\gamma_t)e^{0.5\theta^2\sigma^2} + \gamma_t \left(\alpha_t e^{-\theta\bar{\xi}\sigma} + (1-\alpha_t)e^{\theta\underline{\xi}\sigma} \right)} \right]^2 \right\}. \quad (14)$$

Appendix B Proofs

Proposition 2:

Proof. We start with Equation (12), and let $xr_t := E_t r_{t+1} - r_{f,t}$. Using the approximation $e^x \approx 1+x$, we have

$$xr_t + \frac{1}{2}q_t^2 \approx \eta\theta\sigma q_t(1+x_t + \frac{1}{2}q_t^2) - \frac{\gamma_t}{1-\gamma_t} \left[\alpha_t(1 - \frac{1}{2}\theta^2\sigma^2 - \theta\bar{\xi}\sigma)(xr_t + \bar{\xi}q_t) + (1-\alpha_t)(1 - \frac{1}{2}\theta^2\sigma^2 + \theta\underline{\xi}\sigma)(xr_t - \underline{\xi}q_t) \right].$$

Dropping all second order terms ($\sigma^2, \sigma q_t, q_t^2, \sigma x r_t$) and solving for xr_t yields the result. $\square$

Appendix C Derivation for the Price-Dividend Ratio

First, let us denote the expectation operator $\tilde{E}_t$ for the NEO-EU agent as

$$\tilde{E}_t X_{t+1} = (1-\gamma_t)E_t X_{t+1} + \gamma_t(\alpha_t \bar{X}_{t+1} + (1-\alpha_t)\underline{X}_{t+1}),$$

where $\bar{X}_{t+1}$ and $\underline{X}_{t+1}$ are the perceived best-case (optimistic) and worst-case (pessimistic) scenarios. Then, Euler equation (3)

$$(1-\gamma_t)E_t M_{t+1} R_{t+1} + \gamma_t(\alpha_t \bar{M}_{t+1} \bar{R}_{t+1} + (1-\alpha_t)\underline{M}_{t+1} \underline{R}_{t+1}) = 1,$$

is nothing but $\tilde{E}_t M_{t+1} R_{t+1} = 1$. Now rewriting the return as $R_{t+1} = \frac{PD_{t+1}+1}{PD_t} \frac{D_{t+1}}{D_t}$, and using the discount factor for the CRRA utility function $M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\theta}$, and the assumptions of the Lucas-tree model

$$\log\left(\frac{C_{t+1}}{C_t}\right) = g + \sigma z_{t+1}, \quad \log\left(\frac{D_{t+1}}{D_t}\right) = g + \phi \sigma z_{t+1},$$

we have

$$\tilde{E}_t e^{-\rho - \theta(g + \sigma z_{t+1}) + g + \phi \sigma z_{t+1} + \log(PD_{t+1} + 1) - pd_t} = 1,$$

where $\beta = e^{-\rho}$ and $pd_t = \log(PD_t)$. Next, log-linearizing the expression $\log(PD_{t+1} + 1)$ yields

$$\log(PD_{t+1} + 1) \approx \log(PD + 1) + \omega(pd_{t+1} - pd), \quad \text{where} \quad \omega \equiv PD/(PD + 1) < 1,$$

and PD is the point around which we log-linearized (i.e., the “steady state”). Thus, we have,

$$\tilde{E}_t e^{\psi + \omega pd_{t+1} - pd_t + (\phi - \theta)\sigma z_{t+1}} = 1, \quad \text{with} \quad \psi \equiv -\rho - (\theta - 1)g + \log(PD + 1) - \omega pd.$$

Using $e^x \approx 1 + x$, we have $pd_t = \psi + (\phi - \theta)\sigma \tilde{E}_t z_{t+1} + \omega \tilde{E}_t pd_{t+1}$. Since $\omega \in (0, 1)$, we have a forward looking solution as follows

$$pd_t = \frac{\psi}{1 - \omega} + (\phi - \theta)\sigma \sum_{j=0}^{\infty} \omega^j \tilde{E}_t z_{t+1+j}.$$

One can find the solution by first substituting pd_{t+1} from the expression for pd_t , and then keep iterating forward, where using the law of iterated expectations and $\lim_{j \rightarrow \infty} \omega^j \tilde{E}_t pd_{t+j} = 0$ yields the above equation. At this point we can discuss how internalizing the true dynamics of γ_t and α_t affects the solution of pd_t . Specifically, let γ_t and α_t have an exogenous law of motion. In this case, $\tilde{E}_t z_{t+j+1} = E_t(1 - \gamma_{t+j})z_{t+j+1} + E_t \gamma_{t+j}(\alpha_{t+j}\bar{\xi} - (1 - \alpha_{t+j})\underline{\xi}) = (\bar{\xi} + \underline{\xi})E_t \gamma_{t+j}\alpha_{t+j} - \underline{\xi}E_t \gamma_{t+j}$. Putting a simple autoregressive dynamics on $\gamma_t \alpha_t$ and γ_t with unconditional expectations $\gamma \alpha$ and γ implies that $E_t \gamma_{t+j}\alpha_{t+j} = \gamma \alpha (1 - \rho^j) + \rho^j \gamma_t \alpha_t$ and $E_t \gamma_{t+j} = \gamma (1 - \rho^j) + \rho^j \gamma_t$.²⁶ Thus, we have

$$\tilde{E}_t z_{t+j+1} = (\bar{\xi} + \underline{\xi})(\gamma \alpha (1 - \rho^j) + \rho^j \gamma_t \alpha_t) - \underline{\xi}(\gamma (1 - \rho^j) + \rho^j \gamma_t).$$

Thus, the sum in the pd_t expression above yields

$$pd_t = \psi_0 + \psi_1 h_t, \quad \text{with} \quad h_t \equiv \gamma_t((\bar{\xi} + \underline{\xi})\alpha_t - \underline{\xi}),$$

where $\psi_0 \equiv \frac{\psi}{1 - \omega} + (\phi - \theta)\sigma \left(\frac{1}{1 - \omega} - \frac{1}{1 - \rho\omega} \right) \gamma((\bar{\xi} + \underline{\xi})\alpha - \underline{\xi})$, and $\psi_1 \equiv (\frac{\phi - \theta}{1 - \rho\omega})\sigma$.

²⁶We use the same autocorrelation coefficient ρ for both processes, which is an approximation that follows from α_t having a high autocorrelation and γ_t having a low autocorrelation. Deviating from this assumption does very little, and in general, having more sophisticated exogenous processes makes the final solution less analytically transparent.

REFERENCES

- Abel, Andrew B, 1999, Risk premia and term premia in general equilibrium, *Journal of Monetary Economics* 43, 3–33.
- Allais, Maurice, 1953, Le comportement de l’homme rationnel devant le risque: critique des postulats et axiomes de l’école américaine, *Econometrica* 503–546.
- Baele, Lieven, 2005, Volatility spillover effects in european equity markets, *Journal of Financial and Quantitative Analysis* 40, 373–401.
- Baillie, Richard T, and Ramon P DeGennaro, 1990, Stock returns and volatility, *Journal of financial and Quantitative Analysis* 25, 203–214.
- Baillon, Aurélien, and Han Bleichrodt, 2015, Testing ambiguity models through the measurement of probabilities for gains and losses, *American Economic Journal: Microeconomics* 7, 77–100.
- Baillon, Aurélien, Han Bleichrodt, Umut Keskin, Olivier l’Haridon, and Chen Li, 2018, The effect of learning on ambiguity attitudes, *Management Science* 64, 2181–2198.
- Baillon, Aurélien, and Aysil Emirmahmutoglu, 2018, Zooming in on ambiguity attitudes, *International Economic Review* 59, 2107–2131.
- Baker, Malcolm, and Jeffrey Wurgler, 2006, Investor sentiment and the cross-section of stock returns, *The Journal of Finance* 61, 1645–1680.
- Baker, Scott R, Nicholas Bloom, and Steven J Davis, 2016, Measuring economic policy uncertainty, *The Quarterly Journal of Economics* 131, 1593–1636.
- Bakshi, Gurdip, Dilip Madan, and George Panayotov, 2010, Returns of claims on the upside and the viability of u-shaped pricing kernels, *Journal of Financial Economics* 97, 130–154.
- Bali, Turan G, and Hao Zhou, 2016, Risk, uncertainty, and expected returns, *Journal of Financial and Quantitative Analysis* 51, 707–735.
- Bansal, Ravi, and Amir Yaron, 2004, Risks for the long run: A potential resolution of asset pricing puzzles, *The Journal of Finance* 59, 1481–1509.
- Barro, Robert J, and Gordon Y Liao, 2021, Rare disaster probability and options pricing, *Journal of Financial Economics* 139, 750–769.
- Bekaert, Geert, and Eric Engstrom, 2017, Asset return dynamics under habits and bad environment–good environment fundamentals, *Journal of Political Economy* 125, 713–760.
- Bekaert, Geert, Eric Engstrom, and Andrey Ermolov, 2020, The variance risk premium in equilibrium models, Technical report, National Bureau of Economic Research.

- Bekaert, Geert, and Marie Hoerova, 2014, The vix, the variance premium and stock market volatility, *Journal of econometrics* 183, 181–192.
- Bianchi, Francesco, Cosmin L Ilut, and Martin Schneider, 2018, Uncertainty shocks, asset supply and pricing over the business cycle, *The Review of Economic Studies* 85, 810–854.
- Bollerslev, Tim, George Tauchen, and Hao Zhou, 2009, Expected stock returns and variance risk premia, *The Review of Financial Studies* 22, 4463–4492.
- Bossaerts, Peter, Paolo Ghirardato, Serena Guarnaschelli, and William R Zame, 2010, Ambiguity in asset markets: Theory and experiment, *The Review of Financial Studies* 23, 1325–1359.
- Brandt, Michael W, and Qiang Kang, 2004, On the relationship between the conditional mean and volatility of stock returns: A latent var approach, *Journal of Financial Economics* 72, 217–257.
- Brandt, Michael W, and Leping Wang, 2007, Measuring the time-varying risk-return relation from the cross-section of equity returns.” working paper .
- Brenner, Menachem, and Yehuda Izhakian, 2018, Asset pricing and ambiguity: Empirical evidence, *Journal of Financial Economics* 130, 503–531.
- Buss, Adrian, Lorenzo Schönleber, and Grigory Vilkov, 2019, Expected correlation and future market returns, *Available at SSRN 3114063* .
- Camerer, Colin F, and Howard Kunreuther, 1989, Decision processes for low probability events: Policy implications, *Journal of Policy Analysis and Management* 8, 565–592.
- Campbell, John Y, 1987, Stock returns and the term structure, *Journal of financial economics* 18, 373–399.
- Campbell, John Y, and John H Cochrane, 1999, By force of habit: A consumption-based explanation of aggregate stock market behavior, *Journal of Political Economy* 107, 205–251.
- Campbell, John Y, and Ludger Hentschel, 1992, No news is good news: An asymmetric model of changing volatility in stock returns, *Journal of financial Economics* 31, 281–318.
- Campbell, John Y, and Robert J Shiller, 1988, The dividend-price ratio and expectations of future dividends and discount factors, *The Review of Financial Studies* 1, 195–228.
- Chateauneuf, Alain, Jürgen Eichberger, and Simon Grant, 2007, Choice under uncertainty with the best and worst in mind: Neo-additive capacities, *Journal of Economic Theory* 137, 538–567.
- Cuesdeanu, Horatio, and Jens Carsten Jackwerth, 2018, The pricing kernel puzzle: Survey and outlook, *Annals of Finance* 14, 289–329.
- Dangl, Thomas, and Michael Halling, 2012, Predictive regressions with time-varying coefficients, *Journal of Financial Economics* 106, 157–181.

- Dimmock, Stephen G, Roy Kouwenberg, Olivia S Mitchell, and Kim Peijnenburg, 2015, Estimating ambiguity preferences and perceptions in multiple prior models: Evidence from the field, *Journal of Risk and Uncertainty* 51, 219–244.
- Dimmock, Stephen G, Roy Kouwenberg, Olivia S Mitchell, and Kim Peijnenburg, 2016, Ambiguity aversion and household portfolio choice puzzles: Empirical evidence, *Journal of Financial Economics* 119, 559–577.
- Dow, James, and Sérgio Ribeiro da Costa Werlang, 1992, Uncertainty aversion, risk aversion, and the optimal choice of portfolio, *Econometrica: Journal of the Econometric Society* 197–204.
- Ebert, Sebastian, and Paul Karehnke, 2019, Skewness preferences in choice under risk, *Available at SSRN 3480519* .
- Ebert, Sebastian, and Paul Karehnke, 2021, Skewness preferences in choice under risk, *Available at SSRN* .
- Ellsberg, Daniel, 1961, Risk, ambiguity, and the savage axioms, *The quarterly journal of economics* 643–669.
- Epstein, Larry G, and Tan Wang, 1994, Intertemporal asset pricing under knightian uncertainty, *Econometrica: Journal of the Econometric Society* 283–322.
- Fong, Wai Mun, and Benjamin Toh, 2014, Investor sentiment and the max effect, *Journal of Banking & Finance* 46, 190–201.
- French, Kenneth R, G William Schwert, and Robert F Stambaugh, 1987, Expected stock returns and volatility, *Journal of financial Economics* 19, 3–29.
- Friedman, Milton, and Leonard J Savage, 1948, The utility analysis of choices involving risk, *Journal of Political Economy* 56, 279–304.
- Garcia, Diego, 2013, Sentiment during recessions, *The Journal of Finance* 68, 1267–1300.
- Ghirardato, Paolo, Fabio Maccheroni, and Massimo Marinacci, 2004, Differentiating ambiguity and ambiguity attitude, *Journal of Economic Theory* 118, 133–173.
- Ghysels, Eric, Pedro Santa-Clara, and Rossen Valkanov, 2005, There is a risk-return trade-off after all, *Journal of Financial Economics* 76, 509–548.
- Gilboa, Itzhak, and David Schmeidler, 1989, Maxmin expected utility with non-unique prior, *Journal of Mathematical Economics* 18, 141–153.
- Golez, Benjamin, and Peter Koudijs, 2018, Four centuries of return predictability, *Journal of Financial Economics* 127, 248–263.

- Greenwood, Robin, and Stefan Nagel, 2009, Inexperienced investors and bubbles, *Journal of Financial Economics* 93, 239–258.
- Grevenbrock, Nils, Max Groneck, Alexander Ludwig, and Alexander Zimmer, 2020, Cognition, optimism and the formation of age-dependent survival beliefs, *International Economic Review* 62, 887–918.
- Guidolin, Massimo, and Francesca Rinaldi, 2013, Ambiguity in asset pricing and portfolio choice: A review of the literature, *Theory and Decision* 74, 183–217.
- Guo, Hui, and Robert F Whitelaw, 2006, Uncovering the risk–return relation in the stock market, *The Journal of Finance* 61, 1433–1463.
- Hansen, Lars Peter, and Thomas J Sargent, 2001, Robust control and model uncertainty, *American Economic Review* 91, 60–66.
- Henkel, Sam James, J Spencer Martin, and Federico Nardari, 2011, Time-varying short-horizon predictability, *Journal of financial economics* 99, 560–580.
- Holt, Charles A, and Susan K Laury, 2002, Risk aversion and incentive effects, *American economic review* 92, 1644–1655.
- Holzmeister, Felix, Jürgen Huber, Michael Kirchler, Florian Lindner, Utz Weitzel, and Stefan Zeisberger, 2020, What drives risk perception? a global survey with financial professionals and laypeople, *Management Science* 66, 3977–4002.
- Huang, Dashan, Fuwei Jiang, Jun Tu, and Guofu Zhou, 2015, Investor sentiment aligned: A powerful predictor of stock returns, *The Review of Financial Studies* 28, 791–837.
- Kahneman, Daniel, and Amos Tversky, 1979, Prospect theory: An analysis of decision under risk, *Econometrica* 47, 263–292.
- Kerestecioglu, Semih, Xi Fu, and Charlie X Cai, 2021, Economic uncertainty: Mispricing and ambiguity premium, *Available at SSRN 3655670* .
- Klibanoff, Peter, Massimo Marinacci, and Sujoy Mukerji, 2005, A smooth model of decision making under ambiguity, *Econometrica* 73, 1849–1892.
- Knight, Frank Hyneman, 1921, *Risk, uncertainty and profit*, volume 31 (Houghton Mifflin).
- Kocher, Martin G, Amrei Marie Lahno, and Stefan T Trautmann, 2018, Ambiguity aversion is not universal, *European Economic Review* 101, 268–283.
- Lettau, Martin, and Sydney C Ludvigson, 2010, Measuring and modeling variation in the risk–return trade-off, *Handbook of financial econometrics: Tools and techniques* 617–690.

- Lundblad, Christian, 2007, The risk return tradeoff in the long run: 1836–2003, *Journal of Financial Economics* 85, 123–150.
- Martin, Ian, 2017, What is the expected return on the market?, *The Quarterly Journal of Economics* 132, 367–433.
- Mehra, Rajnish, and Edward C Prescott, 1985, The equity premium: A puzzle, *Journal of monetary Economics* 15, 145–161.
- Merton, Robert C, 1973, An intertemporal capital asset pricing model, *Econometrica: Journal of the Econometric Society* 867–887.
- Merton, Robert C, 1980, On estimating the expected return on the market: An exploratory investigation, *Journal of financial economics* 8, 323–361.
- Miao, Jianjun, Bin Wei, and Hao Zhou, 2019, Ambiguity aversion and the variance premium, *Quarterly Journal of Finance* 9, 1–36.
- Nelson, Daniel B, 1991, Conditional heteroskedasticity in asset returns: A new approach, *Econometrica: Journal of the Econometric Society* 347–370.
- Pástor, L’uboš, Meenakshi Sinha, and Bhaskaran Swaminathan, 2008, Estimating the intertemporal risk–return tradeoff using the implied cost of capital, *The Journal of Finance* 63, 2859–2897.
- Rapach, David E, Jack K Strauss, and Guofu Zhou, 2010, Out-of-sample equity premium prediction: Combination forecasts and links to the real economy, *The Review of Financial Studies* 821–862.
- Rösch, Dominik M, Avanidhar Subrahmanyam, and Mathijs A Van Dijk, 2017, The dynamics of market efficiency, *The review of financial studies* 30, 1151–1187.
- Schlag, Christian, Michael Semenischev, and Julian Thimme, 2021, Predictability and the cross-section of expected returns: A challenge for asset pricing models, *Management Science* 67, 7932–7950.
- Schmeidler, David, 1989, Subjective probability and expected utility without additivity, *Econometrica* 571–587.
- Schmeling, Maik, 2007, Institutional and individual sentiment: smart money and noise trader risk?, *International Journal of Forecasting* 23, 127–145.
- Schreindorfer, David, 2020, Macroeconomic tail risks and asset prices, *The Review of Financial Studies* 33, 3541–3582.
- Segal, Gill, Ivan Shaliastovich, and Amir Yaron, 2015, Good and bad uncertainty: Macroeconomic and financial market implications, *Journal of Financial Economics* 117, 369–397.
- Sichert, Tobias, 2020, The pricing kernel is u-shaped, *Available at SSRN 3095551* .

- Smith, Nicholas JJ, 2014, Is evaluative compositionality a requirement of rationality?, *Mind* 123, 457–502.
- Stambaugh, Robert F, and Yu Yuan, 2017, Mispricing factors, *The Review of Financial Studies* 30, 1270–1315.
- Trautmann, Stefan T, and Gijs Van De Kuilen, 2015, Ambiguity attitudes, *The Wiley Blackwell handbook of judgment and decision making* 1, 89–116.
- Tsai, Jerry, and Jessica A Wachter, 2016, Rare booms and disasters in a multisector endowment economy, *The Review of Financial Studies* 29, 1113–1169.
- Tversky, Amos, and Daniel Kahneman, 1992, Advances in prospect theory: Cumulative representation of uncertainty, *Journal of Risk and Uncertainty* 5, 297–323.
- Wachter, Jessica A, 2013, Can time-varying risk of rare disasters explain aggregate stock market volatility?, *The Journal of Finance* 68, 987–1035.
- Wakker, Peter P, 2010, *Prospect theory: For risk and ambiguity* (Cambridge university press).
- Whitelaw, Robert F, 1994, Time variations and covariations in the expectation and volatility of stock market returns, *The Journal of Finance* 49, 515–541.
- Yu, Jianfeng, and Yu Yuan, 2011, Investor sentiment and the mean–variance relation, *Journal of Financial Economics* 100, 367–381.
- Zhou, Hao, 2018, Variance risk premia, asset predictability puzzles, and macroeconomic uncertainty, *Annual Review of Financial Economics* 10, 481–497.
- Zimper, Alexander, 2012, Asset pricing in a lucas fruit-tree economy with the best and worst in mind, *Journal of Economic Dynamics and Control* 36, 610–628.

Online Appendix.

The following Online Appendix contains the online material referenced in the main document.

Online Appendix.1. Data Sources

The data series and data sources are listed below. For each data series, we use the data for our full sample period (1990-2018) if available. Otherwise, we use all available data contained within our sample period. All series are monthly.

1. **BW**: The Baker and Wurgler (2006) sentiment index.²⁷
2. **C**: The aggregate consumption series is the sum of non-durables and services from the National Income and Product Account (NIPA) monthly series DNDGRC and DSERRC. Population is the U.S. civilian non-institutional population age 16 and over, series CNP16OV at FRED (Federal Reserve Bank of St. Louis).²⁸
3. **CDF**: Empirical CDF of monthly market returns from CRSP (The Center for Research in Security Prices), for returns from 1990 - 2019.
4. **CP**: The option-implied crash probability over a one-month horizon from Martin (2017).²⁹
5. **EP**: The expected equity premium calculated as the difference between the conditional expected market return from the GARCH(1,1) model in Section 3 and the risk-free rate from Kenneth French's data library.³⁰
6. **EPU**: The economic policy uncertainty index from Baker et al. (2016).³¹
7. **IC**: The option-implied market correlation over a one-month horizon from Buss et al. (2019).³²
8. **ICS**: The Index of Consumer Sentiment (ICS) from the University of Michigan.³³
9. **MKTRF**: The realized equity premium from Kenneth French's data library.
10. **$\tilde{M}$** : The right tail of the pricing kernel, based on equation (5) in Sichert (2020), provided to us by Tobias Sichert.
11. **PD**: The price-dividend ratio, constructed as $pd_t = \frac{P_t}{D_t}$ using the data on prices and dividends from Robert Shiller's web page.³⁴

²⁷<http://people.stern.nyu.edu/jwurgler/>

²⁸<https://www.stlouisfed.org>

²⁹<https://personal.lse.ac.uk/martiniw/>

³⁰https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

³¹from the categorical data file at https://www.policyuncertainty.com/us_monthly.html

³²<https://www.vilkov.net/codedata.html>

³³<http://www.sca.isr.umich.edu/>

³⁴http://www.econ.yale.edu/~shiller/data/ie_data.xls

12. **q**: The conditional stock market volatility generated from the GARCH(1,1) model in Section 3.
13. **REC**: NBER recession indicator.³⁵
14. **R_f**: The real risk-free rate, computed as the difference between the (nominal) risk-free rate from Kenneth French's data library and the growth of the consumer price index (CPI).
15. **R_m**: The realized value-weighted market return computed as $\text{MKTRF} + R_f$.
16. **SVIX**: The annualized risk-neutral variance of the excess market return from Martin (2017).³⁶
17. **VIX**: The VIX index from the Chicago Board of Options Exchange. The series was converted from a daily series to a monthly series by taking the last value for each month to be the value for that month.³⁷
18. **VRP**: The variance risk premium constructed based on Martin (2017) as $R_f \text{SVIX}^2 - 12q^2$.

Online Appendix.2. Supplementary Tables

³⁵<https://fred.stlouisfed.org/series/USRECM>

³⁶<https://personal.lse.ac.uk/martiniw/>

³⁷https://www.cboe.com/tradable_products/vix/vix_historical_data/

Table OA.1. Optimism, Ambiguity, Market Crash Risk and Market Correlation

	(1) CP	(2) CP	(3) CP	(4) CP	(5) CP	(6) CP	(7) CP	(8) CP	(9) CP
BW	-0.08** (-2.17)	-0.08** (-2.31)	-0.09 (-1.62)						
HJTZ				-0.13*** (-4.17)	-0.13*** (-4.47)	-0.11** (-2.43)			
ICS							-0.10** (-2.20)	-0.09** (-2.11)	-0.07 (-0.94)
VRP	0.54*** (8.56)			0.55*** (8.69)			0.52*** (8.80)		
VRP _x		0.46*** (11.45)			0.47*** (12.36)			0.45*** (11.57)	
EPU			0.33*** (3.17)			0.34*** (3.14)			0.29*** (2.83)
q	0.53*** (6.81)	0.48*** (6.43)	0.46*** (4.80)	0.58*** (7.38)	0.53*** (7.08)	0.50*** (5.30)	0.51*** (6.75)	0.46*** (6.27)	0.46*** (4.66)
R^2	0.799	0.816	0.603	0.811	0.829	0.608	0.801	0.816	0.596
	(1) IC	(2) IC	(3) IC	(4) IC	(5) IC	(6) IC	(7) IC	(8) IC	(9) IC
BW	-0.29*** (-3.82)	-0.28*** (-3.61)	-0.25*** (-3.21)						
HJTZ				-0.34*** (-4.11)	-0.33*** (-3.94)	-0.29*** (-3.57)			
ICS							-0.31** (-2.46)	-0.33*** (-2.87)	-0.30** (-2.36)
VRP	0.48*** (3.46)			0.51*** (4.20)			0.42*** (2.97)		
VRP _x		0.42*** (3.09)			0.44*** (3.67)			0.36*** (2.68)	
EPU			0.35*** (3.85)			0.38*** (4.30)			0.21** (2.20)
q	0.26*** (3.93)	0.24*** (3.38)	0.23*** (2.72)	0.39*** (5.16)	0.34*** (4.38)	0.32*** (3.74)	0.19** (2.45)	0.16** (2.16)	0.21** (2.52)
R^2	0.478	0.436	0.344	0.509	0.457	0.361	0.472	0.456	0.344

Newey-West t statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table tests predictions 4 and 6. It displays regressions of the option-implied crash probability (CP) for a one-month horizon from Martin (2017) (upper panel) and the option-implied market correlation (IC) for a one-month horizon from (Buss et al., 2019) (lower panel) against a proxy for market uncertainty (the variance risk premium (VRP) based on Martin (2017), a proxy for the variance risk premium computed using VIX (denoted VRP_x), or the economic policy uncertainty (EPU) index from Baker et al. (2016)), a proxy for market optimism (the sentiment index (BW) of Baker and Wurgler (2006), the sentiment index (HJTZ) of Huang et al. (2015), or the Michigan index of consumer sentiment (ICS)), and the conditional stock market volatility (q) estimated from the GARCH(1,1) model. All variables in these regressions are standardized to have a mean of 0 and a standard deviation of 1.

Table OA.2. Return Predictability of the Variance Risk Premium in Low PD Periods

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%
L4.VRP	0.15 (0.43)	-0.17 (-0.47)	0.97** (2.02)	1.01** (2.17)				
L4.VRP _{PD}		1.95** (2.40)		3.45*** (3.98)				
L4.VRP _{REC}			-1.46** (-2.46)	-2.54*** (-4.02)				
L5.VRP					0.66* (1.87)	0.39 (1.04)	0.79 (1.63)	0.81* (1.69)
L5.VRP _{PD}						1.61** (1.98)		2.15** (2.39)
L5.VRP _{REC}							-0.24 (-0.40)	-0.91 (-1.40)
R^2	0.001	0.030	0.032	0.107	0.018	0.038	0.019	0.048
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%	MKTRF%
L4.VRPx	0.06 (0.28)	-0.26 (-1.05)	0.57* (1.75)	0.58* (1.83)				
L4.VRPx _{PD}		1.54*** (3.23)		2.44*** (4.73)				
L4.VRPx _{REC}			-0.85** (-2.16)	-1.72*** (-4.04)				
L5.VRPx					0.38* (1.68)	0.16 (0.65)	0.25 (0.77)	0.25 (0.78)
L5.VRPx _{PD}						1.06** (2.22)		1.16** (2.19)
L5.VRPx _{REC}							0.22 (0.55)	-0.19 (-0.44)
R^2	0.000	0.030	0.014	0.075	0.008	0.022	0.009	0.023

OLS t statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table includes coefficients from predictability regressions with the market excess return multiplied by 100 as the dependent variable. The independent variables (with a lag of four or five months) are the variance risk premium (VRP) computed from SVIX as in Martin (2017), and the variance risk premium computed using VIX instead of SVIX (denoted VRPx), VRP_{PD}, which is the product of VRP and an indicator function that equals 1 for periods in the lowest decile of PD values (VRPx_{PD} is defined analogously to VRP_{PD}), VRP_{REC}, which is the product of VRP and an indicator function that equals 1 for NBER recession periods (VRPx_{REC} is defined analogously to VRP_{REC}). To aid in interpreting the coefficients, VRP and VRPx are each divided by their sample (monthly) standard deviations, 0.00251 and 0.00249, respectively.

Table OA.3. Return Predictability of Crash Risk and Implied Correlation in Low PD Periods

	(1) MKTRF%	(2) MKTRF%	(3) MKTRF%	(4) MKTRF%	(5) MKTRF%	(6) MKTRF%	(7) MKTRF%	(8) MKTRF%
L4.CP	0.66* (1.90)	-0.09 (-0.19)	0.98* (1.80)	0.77 (1.46)				
L4.CP _{PD}		1.40** (2.35)		3.20*** (4.03)				
L4.CP _{REC}			-0.45 (-0.76)	-2.60*** (-3.33)				
L5.CP					0.90** (2.60)	0.53 (1.11)	0.81 (1.50)	0.74 (1.36)
L5.CP _{PD}						0.70 (1.17)		1.14 (1.40)
L5.CP _{REC}							0.13 (0.21)	-0.64 (-0.80)
R^2	0.019	0.046	0.022	0.099	0.034	0.041	0.034	0.044
	(1) MKTRF%	(2) MKTRF%	(3) MKTRF%	(4) MKTRF%	(5) MKTRF%	(6) MKTRF%	(7) MKTRF%	(8) MKTRF%
L4.IC	0.69** (2.53)	0.53* (1.94)	0.77*** (2.82)	0.62** (2.30)				
L4.IC _{PD}		1.03** (2.50)		2.08*** (4.26)				
L4.IC _{REC}			-0.42 (-1.65)	-1.14*** (-3.80)				
L5.IC					0.57** (2.12)	0.44 (1.60)	0.64** (2.32)	0.51* (1.87)
L5.IC _{PD}						0.90** (2.17)		1.73*** (3.50)
L5.IC _{REC}							-0.31 (-1.20)	-0.91** (-2.98)
R^2	0.024	0.047	0.034	0.097	0.017	0.034	0.022	0.066

OLS t statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table includes coefficients from predictability regressions with the market excess return multiplied by 100 as the dependent variable. The independent variables (with a lag of four or five months) are the option-implied crash probability (CP) from Martin (2017), and the option-implied correlation (IC) from Buss et al. (2019), CP_{PD}, which is the product of CP and an indicator function that equals 1 for periods in the lowest decile of PD values (IC_{PD} is defined analogously to CP_{PD}), and CP_{REC}, which is the product of CP and an indicator function that equals 1 for NBER recession periods (IC_{REC} is defined analogously to CP_{REC}). To aid in interpreting the coefficients, CP and IC are each divided by their sample (monthly) standard deviations, 0.011 and 0.132, respectively.

Table OA.4. Return Predictability of the Pricing Kernel U-shape in low PD periods

	(1) MKTRF%	(2) MKTRF%	(3) MKTRF%	(4) MKTRF%	(5) MKTRF%	(6) MKTRF%	(7) MKTRF%	(8) MKTRF%
L4. $\tilde{M}$	0.09 (0.34)	-0.03 (-0.10)	0.14 (0.48)	0.14 (0.49)				
L4. $\tilde{M}_{PD}$		2.95*** (2.79)		3.96*** (3.37)				
L4. $\tilde{M}_{REC}$			-0.26 (-0.46)	-1.17* (-1.92)				
L5. $\tilde{M}$					-0.02 (-0.07)	-0.09 (-0.31)	0.03 (0.12)	0.04 (0.12)
L5. $\tilde{M}_{PD}$						1.65 (1.54)		2.38** (2.00)
L5. $\tilde{M}_{REC}$							-0.30 (-0.54)	-0.85 (-1.38)

OLS t statistics in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: The table includes coefficients from predictability regressions with the market excess return multiplied by 100 as the dependent variable. The independent variables (with a lag of four or five months) is a measure of the increasing portion of the right tail of the pricing kernel, denoted $\tilde{M}$, from equation (5) in Sichert (2020) using the physical distribution from Bekaert and Hoerova (2014). The variable $\tilde{M}_{PD}$ is the product of $\tilde{M}$ and an indicator function that equals 1 for periods in the lowest decile of PD values and $\tilde{M}_{REC}$, which is the product of $\tilde{M}$ and an indicator function that equals 1 for NBER recession periods. To aid in interpreting the coefficients, $\tilde{M}$ is divided by its standard deviation, 4.67.

Table OA.5. GARCH(1,1) specifications of market return for up to 12 lags of the *pd* ratio.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
L.pd	-3.3e-04 (0.00016)											
L2.pd		-3.3e-04 (0.00016)										
L3.pd			-3.2e-04 (0.00016)									
L4.pd				-3.2e-04 (0.00016)								
L5.pd					-3.0e-04 (0.00016)							
L6.pd						-3.1e-04 (0.00016)						
L7.pd							-2.9e-04 (0.00016)					
L8.pd								-3.0e-04 (0.00016)				
L9.pd									-3.1e-04 (0.00016)			
L10.pd										-3.1e-04 (0.00016)		
L11.pd											-2.9e-04 (0.00016)	
L12.pd												-3.0e-04 (0.00015)
Constant	2.6e-02 (0.008)	2.6e-02 (0.009)	2.6e-02 (0.008)	2.6e-02 (0.008)	2.5e-02 (0.008)	2.5e-02 (0.008)	2.4e-02 (0.008)	2.4e-02 (0.008)	2.5e-02 (0.008)	2.5e-02 (0.008)	2.4e-02 (0.008)	2.4e-02 (0.008)
ARCH												
L.arch	1.8e-01 (0.048)	1.8e-01 (0.047)	1.8e-01 (0.048)	1.8e-01 (0.047)	1.8e-01 (0.048)	1.8e-01 (0.048)	1.8e-01 (0.048)	1.8e-01 (0.048)	1.8e-01 (0.047)	1.8e-01 (0.048)	1.8e-01 (0.047)	1.8e-01 (0.047)
L.garch	7.8e-01 (0.054)	7.8e-01 (0.053)	7.8e-01 (0.053)	7.8e-01 (0.053)	7.8e-01 (0.054)	7.8e-01 (0.054)	7.8e-01 (0.054)	7.8e-01 (0.054)	7.8e-01 (0.053)	7.8e-01 (0.053)	7.8e-01 (0.053)	7.8e-01 (0.053)
Constant	9.0e-05 (0.00005)	9.0e-05 (0.00004)	9.0e-05 (0.00005)	9.0e-05 (0.00005)	9.0e-05 (0.00005)	9.1e-05 (0.00005)	9.2e-05 (0.00005)	9.2e-05 (0.00005)	9.1e-05 (0.00004)	9.2e-05 (0.00005)	9.1e-05 (0.00004)	9.2e-05 (0.00004)
N	348	347	346	345	344	343	342	341	340	339	338	337

Standard errors in parentheses.

Online Appendix.3. Inequalities for Decision Anomalies

Three of the main systematic deviations from EU include two empirical violations of the EU independence axiom (the common consequence effect (Allais, 1953; Kahneman and Tversky, 1979), and the common ratio effect (Kahneman and Tversky, 1979)) and systematic deviations from risk aversion (the fourfold pattern of risk preferences) documented by Tversky and Kahneman (1992) which provides evidence that a lottery's skewness has a first-order effect on decision making, a finding that cannot be captured by EU (Ebert and Karehnke, 2019). These behaviors can be explained by the NEO-EU agent with the same calibrated values of α_t and γ_t that match the aggregate market moments.

We consider a NEO-EU agent with log utility of consumption that spends \$5,000 in monthly consumption expenditures.³⁸ We provide the inequalities that are satisfied by this agent which resolve the decision anomalies. Setting $\alpha = 0.138$ and $\gamma = 0.036$ to be consistent with the asset pricing moments in Section 3, the agent exhibits the common consequence effect, preferring \$2400 with certainty over a lottery offering a 33% chance of \$2500, a 66% chance of \$2400, and a 1% chance of \$0, while preferring a 33% chance of \$2500 over a 34% chance of \$2400. The same consumer exhibits the common ratio effect from Kahneman and Tversky (1979), preferring \$3000 with certainty over an 80% chance of \$4000, but preferring a 20% chance of \$4000 over a 25% chance of \$3000. In addition, the consumer exhibits the fourfold pattern of risk preferences (Tversky and Kahneman, 1992) in which the consumer prefers (i) a loss of \$5 with certainty over a 5% chance of losing \$100; (ii) a gain of \$95 with certainty over a 95% chance of winning \$100; (iii) a 5% chance of winning \$100 over \$5 with certainty; and (iv) a 95% chance of losing \$100 over losing \$95 with certainty. Choices (i) and (iii) roughly resemble the purchase of an insurance policy and a lottery ticket, respectively. These three anomalies (common consequence effect, common ratio effect, and fourfold pattern) imply inequalities OA.1 - OA.8 which must hold if the NEO-EU agent rationalizes these behaviors. We find that all eight inequalities hold simultaneously for the NEO-EU agent under our calibration.

Online Appendix.3.1. The Common Consequence Effect

The NEO-EU agent exhibits the common consequence effect as observed by Kahneman and Tversky (1979) if the following inequalities hold:

$$\begin{aligned} & \log(C + \$2400) \\ & > ((1 - \alpha)\gamma + 0.01(1 - \gamma)) \log(C) + (\alpha\gamma + 0.66(1 - \gamma)) \log(C + \$2500) + 0.33(1 - \gamma) \log(C + \$2400). \end{aligned} \tag{OA.1}$$

$$\begin{aligned} & ((1 - \alpha)\gamma + 0.66(1 - \gamma)) \log(C) + (\alpha\gamma + 0.34(1 - \gamma)) \log(C + \$2400) \\ & < ((1 - \alpha)\gamma + 0.67(1 - \gamma)) \log(C) + (\alpha\gamma + 0.33(1 - \gamma)) \log(C + \$2500). \end{aligned} \tag{OA.2}$$

³⁸This is consistent with the average monthly consumer expenditures in the U.S., which was \$5253 in 2019 (<https://www.bls.gov/news.release/cesan.nr0.htm>).

Online Appendix.3.2. The Common Ratio Effect

The NEO-EU agent exhibits the common ratio effect as observed by Kahneman and Tversky (1979) if the following inequalities hold:

$$\log(C + \$3000) > ((1 - \alpha)\gamma + 0.2(1 - \gamma)) \log(C) + (\alpha\gamma + 0.8(1 - \gamma)) \log(C + \$4000). \quad (\text{OA.3})$$

$$\begin{aligned} & ((1 - \alpha)\gamma + 0.75(1 - \gamma)) \log(C) + (\alpha\gamma + 0.25(1 - \gamma)) \log(C + \$3000) \\ & < ((1 - \alpha)\gamma + 0.8(1 - \gamma)) \log(C) + (\alpha\gamma + 0.2(1 - \gamma)) \log(C + \$4000). \end{aligned} \quad (\text{OA.4})$$

Online Appendix.3.3. The Fourfold Pattern of Risk Preferences

The NEO-EU agent exhibits the fourfold pattern of risk preferences as observed by Tversky and Kahneman (1992) if the following inequalities hold:

$$\log(C + 5) < ((1 - \alpha)\gamma + 0.95(1 - \gamma)) \log(C) + (\alpha\gamma + 0.05(1 - \gamma)) \log(C + 100). \quad (\text{OA.5})$$

$$\log(C - 95) < ((1 - \alpha)\gamma + 0.95(1 - \gamma)) \log(C - 100) + (\alpha\gamma + 0.05(1 - \gamma)) \log(C). \quad (\text{OA.6})$$

$$\log(C + 95) > ((1 - \alpha)\gamma + 0.05(1 - \gamma)) \log(C) + (\alpha\gamma + 0.95(1 - \gamma)) \log(C + 100). \quad (\text{OA.7})$$

$$\log(C - 5) > ((1 - \alpha)\gamma + 0.05(1 - \gamma)) \log(C - 100) + (\alpha\gamma + 0.95(1 - \gamma)) \log(C). \quad (\text{OA.8})$$